THE BOLZA SURFACE IS A STRICT LOCAL MAXIMUM OF THE FIRST LAPLACE EIGENVALUE ON GENUS-TWO TEICHMÜLLER SPACE

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Abstract. Let B be the Bolza surface, the most symmetric closed hyperbolic surface of genus two. Strohmaier and Uski conjectured in 2013 that B maximises the first positive Laplace eigenvalue λ_1 among closed hyperbolic surfaces of genus two, and local maximality has been asserted on numerical grounds. We give a computer-assisted proof that B is a *strict local maximum* of λ_1 on the Teichmüller space $\mathcal{T}_2$. The proof does not depend on the eigenvalue enclosure of Strohmaier and Uski: we certify $\lambda_1(B) \geq 3.8388757010$ ourselves by the Lehmann–Goerisch method with an $H(\mathrm{div})$-conforming flux on the glued octagon. The proof is conditional on quoted results: two lemmas and geodesic data of Fortier Bourque and Petri, the Selberg trace formula, and the Lehmann–Goerisch theorem. Because $\lambda_1(B)$ has multiplicity three, λ_1 is not differentiable at B. We analyse instead the analytic 3×3 matrix M that carries the eigenvalue cluster. Its first-order part A is traceless by symmetry and has rank exactly three. Its kernel K is a three-dimensional absolutely irreducible module for $\mathrm{Isom}(B)$, identified exactly as the (-1)-eigenspace of the antipodal isometry. On K the Hessian of the cluster sum is a scalar multiple of an invariant inner product, and a certified Ky Fan bound along one explicit deformation shows that this scalar is negative. The certified ingredients are a Selberg-trace-formula count, conforming finite elements on the glued octagon in ball arithmetic (Rayleigh–Ritz upper bounds and a Lehmann–Goerisch lower bound), exact character computations in $\mathbb{Q}(\zeta_{24})$, and certified rank computations. The finite-dimensional logic of the proof has been checked in Lean 4, with the analytic and computational inputs stated as explicit hypotheses.

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Date: Draft, October 10, 2026.

2020 *Mathematics Subject Classification.* 58J50 (primary); 32G15, 35P15, 65G30, 65N30 (secondary).

Key words and phrases. Bolza surface, Laplace eigenvalues, hyperbolic surfaces, Teichmüller space, eigenvalue optimisation, computer-assisted proof, interval arithmetic.

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1. Introduction

1.1. **The problem.** For a closed hyperbolic surface X let $0 = \lambda_0 < \lambda_1(X) \leq \lambda_2(X) \leq \cdots$ be the spectrum of the Laplace–Beltrami operator. In genus two the *Bolza surface* B, obtained by gluing opposite sides of the regular hyperbolic octagon with angles $\pi/4$, is distinguished in several ways. It has the largest isometry group (of order 96) and the longest systole [35, 34], and, conjecturally, the largest first eigenvalue.

1.2. **History.** Jenni [21] estimated $\lambda_1(B) \in [3.83, 3.85]$ and claimed that its multiplicity is 3; Cook [7] found a gap in the multiplicity argument (both as reported in [34, 15]). Strohmaier and Uski [34, §5] computed

$$\lambda_1(B) \approx 3.83887258842199518586622450435464597081915015 7,$$

stating that "the rigorous implementation that uses very simple bounds in its current form gives a rigorous error bound of 10^{-6}". They wrote: "we conjecture that this is the maximal value of the first eigenvalue of the Laplace operator for constant negative curvature -1 in genus 2. This

conjecture is supported by our calculations in Teichmüller space (the corresponding analysis will be discussed elsewhere)". We have not found the promised analysis in print.

In his lecture notes, Strohmaier [35, pp. 22–23] made the local statement explicit: "Numerical evidence suggests that this is the global maximum for constant negative curvature genus 2 surfaces. The reason for it being locally maximal is however its degeneracy. [...] Numerically one can see that no matter what perturbation one chooses, none of the eigenvalues λ_1, λ_2 and λ_3 has an extremal value at the Bolza surface." Cook [7, Conj. 5.1] restated the global conjecture and proposed a route through local maximality: "It is thought that by analyzing the nodal lines of eigenfunctions relating to χ_8 under perturbations in Teichmüller space, one would be able to show that the Bolza surface is a local maximum for λ_1."

Upper bounds now come close to $\lambda_1(B)$. Kravchuk, Mazáč and Pal [27, Thm. 1.1] proved, with exact rational verification, that every hyperbolic orbifold of genus 2 satisfies $\lambda_1 < 3.8388976481$. This is 1.04×10^{-5} above $\lambda_1(B)$, and the authors call it "very strong evidence for the conjecture that the Bolza surface maximizes λ_1 among all $g = 2$ orbifolds". Bonifacio [3, 2] obtained the numerical bootstrap bound $\lambda_1 \le 3.8388977$. Fortier Bourque and Petri proved that $\lambda_1(B)$ has multiplicity 3 [15, Thm. 7.1], completing Cook's work, and developed linear programming bounds for hyperbolic surfaces [14]; they describe B as the "conjectured maximizer" in genus 2. Radcliffe [33] gives numerical evidence that the single-correlator bootstrap bound does not reach $\lambda_1(B)$ even in the limit ("$\tilde{\lambda}_1^{\infty} - \lambda_1 \ge \Delta_{\infty} \approx 5.59683 \times 10^{-6}$"), so the bootstrap route alone does not close the global problem as it stands. As of 2026 the global conjecture is still described as open [19, 14, 2]; see also [32, 18].

To our knowledge, local maximality has not been proved anywhere. Since global maximality is open, it cannot be deduced from a global result. The rigorous bounds [27, 34] show only that no genus-two surface exceeds $\lambda_1(B)$ by more than about 1.1×10^{-5}; they give no local information.

1.3. **Main result.** We write $\lambda_* := \lambda_1(B)$, E for the λ_*-eigenspace, $G := \mathrm{Isom}(B)$ and $\Lambda := \lambda_1 + \lambda_2 + \lambda_3$. The tangent space $T_B\mathcal{T}_2$ is identified with the space of transverse-traceless (TT) tensors $\{\mathrm{Re}\,\varphi : \varphi \in Q(B)\}$, where $Q(B)$ is the space of holomorphic quadratic differentials (Section 2).

Hypothesis 1.1 (SU; optional, not used in the proof)**.** The Bolza surface has a Laplace eigenvalue in the interval [3.83888625884, 3.83888825884] [34, §5].

An earlier version of this proof assumed Hypothesis 1.1 for its lower bound on $\lambda_1(B)$. It is now replaced by the certified bound of Section 7.3, and Hypothesis 1.1 is used only as a cross-check and for alternative (sharper) constants, which we record alongside.

Main Theorem. Assume the quoted results listed in Table 1, namely [15, Lemmas 7.2–7.3], the geodesic data of [15, §7.4], the Selberg trace formula in the normalisation of [15] with [15, Lemma 3.2], the Lehmann–Goerisch theorem [28, Thms. 4.1, 4.3] (see also [1, Thm. 2.1] as reproduced in [37]), and the standard facts listed there. Then B is a strict local maximum of λ_1 on $\mathcal{T}_2$: there is a neighbourhood N of B in $\mathcal{T}_2$ such that $\lambda_1(X) < \lambda_1(B)$ for every $X \in N \setminus \{B\}$. More precisely:

(i) $\lambda_1(B)$ has multiplicity 3, and $\lambda_1(B) \in [3.8388757010,\ 3.8388872830]$.

(ii) The first-order cluster map $A\colon T_B\mathcal{T}_2 \to \mathrm{Sym}_0(E)$ has rank 3. Its kernel K is 3-dimensional and is an absolutely irreducible G-module; it equals the (-1)-eigenspace of the antipodal isometry.

(iii) Off K, λ_1 decreases to first order: $\lambda_1 - \lambda_1(B) \le -c_A|w|/2$ in the regime of case (i) of Lemma 6.1. On K, the Hessian of Λ satisfies $D^2\Lambda|_K = c \cdot g_{\mathrm{inv}}$ for a G-invariant inner product g_{inv} and a constant $c < 0$.

(iv) Along the explicit equiangular family F_B (Section 4.4), $(\Lambda\circ x)''(0) \le -9.3411$; the central value of the certified enclosure is -14.274.

Both ends of the enclosure in (i) are our own certificates: the upper end by conforming Rayleigh–Ritz, the lower end by Lehmann–Goerisch (Section 7.3). The lower bound uses only the trace-formula count $\lambda_4 > 4.0$ and the Lehmann–Goerisch theorem; it does not use the multiplicity lemmas of [15], which are still needed for multiplicity 3. The second-order certificate needs $\lambda_1(B) \ge 3.8388029653$ and the first-order certificate $\lambda_1(B) \ge 3.83788$; the certified 3.8388757010 exceeds both. If Hypothesis 1.1 is assumed in addition, the constant in (iv) improves to -12.8504 (Remark 10.1). The proof also uses published results that we quote rather than reprove (the Selberg trace formula in the normalisation of [15], the geodesic data and two lemmas of [15] excluding low-dimensional representations in the λ_1-eigenspace, the Lehmann–Goerisch theorem, the identification of $T_B\mathcal{T}_2$ with TT tensors, and the Fischer–Marsden splitting); they are listed in Table 1.

1.4. **Method.** The function λ_1 is not differentiable at B, because $\lambda_1(B)$ is triple. The proof has four steps.

Cluster matrix. Near B the three lowest positive eigenvalues are the eigenvalues of a real-analytic symmetric 3×3 matrix M with $M(0) = \lambda_* I$ (Theorem 3.1). Write $M(v) = \lambda_* I + A(v) + Q_2(v) + O(|v|^3)$. The first-order term $A(v) = [b_v(e_i, e_j)]$ is intrinsic on $T_B\mathcal{T}_2$ and G-equivariant (Lemma 3.3).

First order. By symmetry $\operatorname{tr} A \equiv 0$ (Proposition 5.1). A certified computation shows $A \neq 0$ (Proposition 5.2). Exact representation theory then forces $\ker A = K_0$, the (-1)-eigenspace of the antipodal isometry, with $\operatorname{rank} A = 3$ (Proposition 4.4).

Second order. A local-maximum criterion for degenerate clusters (Lemma 6.1) reduces the theorem to negative definiteness of $D^2\Lambda$ on K. Because K is absolutely irreducible, $D^2\Lambda|_K = c\, g_{\mathrm{inv}}$ once $D^2\Lambda$ is known to be G-invariant in a chart where Λ is C^2 (Lemma 6.4). So a single direction in K decides the sign of c. We take the tangent w_B of the equiangular family F_B, which lies in K exactly (Lemma 4.7). A Ky Fan trace bound with a certified trial space gives $(\Lambda \circ x)''(0) \le -9.3411$ (Proposition 6.8).

Chart. Six explicit trace functions are certified to be local coordinates of $\mathcal{T}_2$ at B (Proposition 6.2). This gives an analytic 6-dimensional slice $\mathcal{S}$ of the octagon space on which all of the above holds and which maps homeomorphically onto a neighbourhood of B.

1.5. **What is computer-assisted.** The following steps are certified by computer. All are done in ball arithmetic (Arb, through python-flint 0.9.0) or in exact rational arithmetic (sympy 1.14.0, exact arithmetic in $\mathbb{Q}(\zeta_{24})$). Floating-point linear algebra is used only to *propose* candidates (test functions, eigenvectors, correctors), which are then verified rigorously.

(a) Isolation of the cluster: a Selberg trace formula count (Section 7.1).
(b) Upper bounds for $\lambda_3(B)$ by conforming finite elements on the glued octagon (Section 7.2), and a lower bound for $\lambda_1(B)$ by the Lehmann–Goerisch method with $H(\mathrm{div})$-conforming fluxes (Section 7.3).
(c) $A \neq 0$ (Section 7.5).
(d) The second-order sign bound along F_B (Section 7.6).
(e) The rank of the constraint Jacobian of the octagon space, and the rank of the differential of the map to $\mathcal{T}_2$ (Section 7.7).
(f) The characters of G on $T_B\mathcal{T}_2$ and the decomposition $T_B\mathcal{T}_2 = K_0 \oplus W_0$ (Section 4; exact).

Everything else is a written argument.

1.6. **Related work.** The pattern "symmetry gives criticality, a computation gives non-degeneracy" is familiar from systolic geometry. Fortier Bourque, Martínez-Granado and Vargas Pallete [16] proved that the extremal-length systole has a strict local maximum at B, combining eutaxy (from symmetry) with perfection (from a rank computation); see also [17]. In our setting the analogue of perfection fails: $\operatorname{rank} A = 3 < 6 = \dim \mathcal{T}_2$, so a genuine second-order certificate on the kernel is needed. For flat tori, local maximality of eigenvalue functionals is classical (see [23, §3.3]). For the spectral determinant, B is a critical point by symmetry [22], and numerically a local maximum [34, §7.1]. The area-normalised problem over *all* metrics is different: there the sharp genus-two constant 16π is attained by a singular metric conformal to B [31, 20] (see also [25]), and the second variation for that problem was only recently written down [24]. For hyperbolic surfaces that bound gives only $\lambda_1 \le 4$ [15, §6].

Our result is consistent with Strohmaier's numerical assertion of local maximality, but it qualifies the stated mechanism. Along directions in the 3-dimensional kernel K (for example along F_B) all three eigenvalue branches are stationary to first order, so the statement that "none of the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ has an extremal value" fails there; it holds on the open dense set of directions off K (Remark 10.2).

1.7. **Organisation.** Section 2 sets up the octagon space and the pulled-back forms. Section 3 proves analyticity of the cluster and the intrinsic nature of A. Section 4 contains the exact representation theory. Sections 5 and 6 carry out the first- and second-order analysis. Section 7 describes the certified computations, including the lower bound for $\lambda_1(B)$, and Section 8 assembles the proof. Section 9 describes a Lean formalisation of the finite-dimensional logic, and Section 10 collects remarks and open problems. Appendix A tabulates the certificate data and Appendix B explains how to reproduce them.

2. Setup

2.1. **Hyperbolic plane.** We use the hyperboloid model $\mathbb{H}^2 = \{X \in \mathbb{R}^3 : \mathsf{B}(X,X) = 1,\ X_0 > 0\}$ with $\mathsf{B} = \operatorname{diag}(1,-1,-1)$, and the associated Klein (projective) coordinates. Orientation-preserving isometries form $SO^+(2,1) \cong PSL(2,\mathbb{R})$.

2.2. **The octagon space.** Let $x = (V_0, \dots, V_7) \in \mathbb{R}^{16}$ be the Klein coordinates of the vertices of a convex hyperbolic octagon, with side k joining V_k and V_{k+1} (indices mod 8). The Bolza gluing pairs side $k+4$ with side k ($k = 0, \dots, 3$) by the orientation-preserving isometry $g_k(x) \in SO^+(2,1)$ with $V_{k+4} \mapsto V_{k+1}$ and $V_{k+5} \mapsto V_k$. Explicitly, $g_k = F(V_{k+1}, V_k)\, F(V_{k+4}, V_{k+5})^{-1}$, where $F(P,Q) = [P, e, n]$ is the Lorentz-orthonormal frame with $e \propto Q - \mathsf{B}(P,Q)P$ and $n \propto \mathsf{B}(P \times e)$.

Let $\mathcal{C} \subset \mathbb{R}^{16}$ be the set of x satisfying the five constraints

$$L_{k+4} = L_k \ (k = 0, \dots, 3), \qquad \textstyle\sum_k \theta_k = 2\pi,$$

where L_k are the side lengths and θ_k the interior angles. By the Poincaré polygon theorem, each $x \in \mathcal{C}$ yields a closed hyperbolic surface B_x of genus 2, with the single vertex cycle $0 \to 5 \to 2 \to 7 \to 4 \to 1 \to 6 \to 3 \to 0$. The point $x_B \in \mathcal{C}$ is the regular octagon, with sides $2\operatorname{arccosh}(1+\sqrt{2}) = 3.0571418389620$ and angles $\pi/4$; $B = B_{x_B}$. By the certified rank of the constraint Jacobian (Proposition 7.6) and the analytic implicit function theorem, $\mathcal{C}$ is a real-analytic 11-manifold near x_B, with $T_{x_B}\mathcal{C} = \ker J$.

2.3. **The map to Teichmüller space.** The surface B_x is marked through the isomorphism $\Gamma_{x_B} \to \Gamma_x$, $g_k(x_B) \mapsto g_k(x)$; this marking is induced by the mesh homeomorphism Φ_x below and is constant in x up to isotopy. This defines $\pi \colon \mathcal{C} \to \mathcal{T}_2$. The map π is continuous, because it factors through the analytic holonomy map $x \mapsto (g_k(x))_k$ and the quotient by conjugation.

2.4. **The mesh map and the pulled-back forms.** The octagon is triangulated by a fixed fan of geodesic triangles. Nodes on the sides are placed at exact hyperbolic-arclength fractions j/n and matched $j \leftrightarrow n-j$ by the side pairings. Interior nodes are given by explicit isometry-equivariant formulas: the centre is the normalised vertex sum, and interpolation uses the symmetric hyperbolic-arclength formula $\sinh(sL)/\sinh L$. The parameter n is the mesh resolution.

On a triangle with hyperboloid vertices P_0, P_1, P_2 set

$$W(a,b) = aP_0 + bP_1 + (1-a-b)P_2, \qquad X(a,b) = W/\sqrt{\mathsf{B}(W,W)}.$$

Cellwise, these maps define a piecewise-analytic, bi-Lipschitz homeomorphism $\Phi_x \colon B \to B_x$ that is compatible with the gluings. Pulling back the Dirichlet and L^2 forms of B_x gives, on $H^1(B)$,

$$q_x(f) = \sum_T \int (\nabla f)^{\mathsf{T}} a_T \nabla f, \qquad m_x(f) = \sum_T \int \rho_T f^2, \tag{1}$$

with

$$a_T = \frac{\operatorname{adj}(LL^{\mathsf{T}} - QG)}{D\sqrt{Q}}, \qquad \rho_T = D\,Q^{-3/2},$$

where $Q = \mathsf{B}(W,W) \geq 1$, $W_1 = P_0 - P_2$, $W_2 = P_1 - P_2$, $L_i = \mathsf{B}(W_i, W)$, $G_{ij} = \mathsf{B}(W_i, W_j)$ and $D = |\det[P_0 P_1 P_2]|$. (These formulas were derived by hand and checked against a finite-difference pullback; the certified area enclosure $\to 4\pi$ is a further check.) The pencil (q_x, m_x) has the Laplace spectrum of B_x.

2.5. **Notation.**

- E is the λ_*-eigenspace of B; e_1, e_2, e_3 is an m_0-orthonormal basis.
- $G = \operatorname{Isom}(B)$ has order 96; $G^+ = \operatorname{Isom}^+(B) \cong GL(2,3)$; $J = r^4$ is the hyperelliptic involution.
- $T_B := T_B\mathcal{T}_2 \cong \{\text{TT tensors on } (B, h_0)\} = \{\operatorname{Re}\varphi : \varphi \in Q(B)\}$, where h_0 is the hyperbolic metric of B ([29, §1], citing [36]; see also [8, §2]). As a real G-module, G^+ acts $\mathbb{C}$-linearly by pull-back and orientation-reversing elements act $\mathbb{C}$-antilinearly.
- Octagon symmetries: r is the rotation by $\pi/4$ (relabelling $k \mapsto k+1$), acting on $\mathbb{R}^{16}$ by $(rx)_k = \operatorname{Rot}_{\pi/4}(x_{k-1})$; s_{mid} is the reflection in an axis through two opposite side midpoints; s_{vert} is the reflection in a diagonal through two opposite vertices.
- In the model $y^2 = x^5 - x$ (Section 4): t is an element of order 3, $u \colon x \mapsto i/x$, $\kappa \colon (x,y) \mapsto (\bar{x}, \bar{y})$, and a is the antipodal isometry, the order-4 orientation-reversing lift of $x \mapsto -1/\bar{x}$.

3. Perturbation of the eigenvalue cluster

3.1. **Analyticity of the cluster.** A *pulled-back family* is a map $t \mapsto (a_t, \rho_t) \in L^\infty$, $t \in U \subset \mathbb{R}^d$, of measurable coefficient fields on B (written in the reference simplices of the mesh), defining forms $q_t(f,g) = \int a_t \nabla f \cdot \nabla g$ and $m_t(f,g) = \int \rho_t f g$ on $H := H^1(B)$. We assume:

- (H1) $t \mapsto (a_t, \rho_t) \in L^\infty$ is C^k, $k \in \{2, 3, \dots, \infty, \omega\}$; for $k = \omega$ it extends holomorphically to a complex neighbourhood of 0;
- (H2) $a_0 \geq \alpha I$ and $\rho_0 \geq \alpha > 0$ almost everywhere;
- (H3) for real t, (q_t, m_t) is the Dirichlet/L^2 pair of a closed hyperbolic surface B_t pulled back by a bi-Lipschitz homeomorphism;
- (H4) $\operatorname{Spec}(B) \cap [\lambda_* - r_0, \lambda_* + r_0] = \{\lambda_*\}$ with multiplicity 3.

The octagon family on $\mathcal{C}$ satisfies (H1)–(H3) with $k = \omega$: a_T and ρ_T are analytic in the vertex coordinates, at x_B we have $Q \geq 1$ and $D > 0$ uniformly over the finitely many elements, and the interpolation $\sinh(sL)/\sinh L$ is analytic in $\cosh L$. The same holds on every analytic sub-family. Hypothesis (H4) holds with $r_0 = 0.16$ by Lemma 7.2.

Theorem 3.1 (Cluster reduction)**.** *Assume (H1)–(H4) and let* $\Gamma = \{|z - \lambda_*| = r_0/2\}$*. For* t *in a neighbourhood of* 0*:*

(1) exactly $\lambda_1(t), \lambda_2(t), \lambda_3(t)$ *lie inside* Γ*, and none lies on* Γ*;*
(2) the Riesz projection $P(t) = (2\pi i)^{-1} \oint_\Gamma (zM_t - Q_t)^{-1} M_t \, dz$ *is* C^k *(holomorphic if* $k = \omega$*), and for real* t *it is the* m_t*-orthogonal projection onto the cluster;*
(3) there is a C^k *symmetric matrix* $M(t) \in \mathrm{Sym}(3)$ *whose eigenvalues are* $\lambda_1(t), \lambda_2(t), \lambda_3(t)$*; in particular* $\Lambda = \operatorname{tr} M$ *is* C^k *(real-analytic if* $k = \omega$*);*
(4) $M(0) = \lambda_* I$ *and* $dM(0)[v] = A(v) := [b_v(e_i, e_j)]_{ij}$*, where* $b_v = \partial_v q - \lambda_* \partial_v m$*;*
(5) $M(v) = \lambda_* I + A(v) + Q_2(v) + R(v)$ *with* Q_2 *quadratic and* $|R(v)| \le C|v|^3$ *(*$k \ge 3$*) or* $R(v) = o(|v|^2)$ *(*$k = 2$*), uniformly on a ball.*

Proof. Step 1. Let $Q_t, M_t \colon H_{\mathbb{C}} \to H'_{\mathbb{C}}$ be the operators of q_t, m_t (bilinear pairing). The maps $a \mapsto Q[a]$, $\rho \mapsto M[\rho]$ are bounded linear from L^∞, so $t \mapsto Q_t, M_t$ inherit the regularity of (H1). M_t is compact, since it factors through L^2 (Rellich).

Step 2. For real t near 0, $q_t + m_t$ is an inner product equivalent to the H^1 norm, and $S_t = (Q_t + M_t)^{-1} M_t$ is compact, self-adjoint and positive for it. The spectral theorem gives an m_t-orthonormal eigenbasis (φ_k) with eigenvalues $\lambda_k(t) \to \infty$, which by (H3) are the eigenvalues of B_t, and $(zM_t - Q_t)^{-1} M_t f = \sum_k m_t(f, \varphi_k)(z - \lambda_k)^{-1} \varphi_k$ for $z \notin \{\lambda_k\}$.

Step 3. By (H4), $zM_0 - Q_0$ is invertible for $z \in \Gamma$ with uniformly bounded inverse. By the Neumann series, applied uniformly on the compact contour, $zM_t - Q_t$ is invertible for $(t, z) \in U' \times \Gamma$, and the resolvent is C^k in t uniformly in z.

Step 4. Differentiating under the contour integral shows that $P(t)$ is C^k. For real t the expansion of Step 2 identifies $P(t)$ with the m_t-orthogonal projection onto the eigenvectors inside Γ. Idempotents at distance < 1 have equal rank, so $\operatorname{rank} P(t) = 3$. Applying the same argument to a contour Γ' enclosing $[-\frac{1}{2}, \lambda_* + r_0/2]$, which at $t = 0$ contains exactly $\{0, \lambda_*, \lambda_*, \lambda_*\}$ (here we use $\mathrm{Spec}(B) \cap (0, 3.3] = \emptyset$), shows that exactly four eigenvalues lie below $\lambda_* + r_0/2$. One is 0, so the three inside Γ are $\lambda_1, \lambda_2, \lambda_3$.

Step 5. Put $f_i(t) = P(t)e_i$, $G(t)_{ij} = m_t(f_i, f_j)$ (so $G(0) = I$), $\tilde{f} = f\, G^{-1/2}$ and $M(t)_{ij} = q_t(\tilde{f}_i, \tilde{f}_j)$. Then M is C^k and symmetric, and for real t, $(\tilde{f}_j)$ is an m_t-orthonormal basis of $\operatorname{Ran} P(t)$, so the eigenvalues of $M(t)$ are $\lambda_1, \lambda_2, \lambda_3$.

Step 6. Since $\tilde{f}_i(0) = e_i$ and $q_0(g, e_j) = \lambda_* m_0(g, e_j)$, differentiating M_{ij} and the identity $m_t(\tilde{f}_i, \tilde{f}_j) = \delta_{ij}$ gives $M'_{ij} = q'(e_i, e_j) - \lambda_* m'(e_i, e_j) = b_v(e_i, e_j)$. Item 5 is Taylor's theorem. □

Remark 3.2. Only the individual eigenvalues can fail to be differentiable in several parameters (compare the eigenvalues $\pm|t|$ of $\left(\begin{smallmatrix} t_1 & t_2 \\ t_2 & -t_1 \end{smallmatrix}\right)$); the total projection and every symmetric function of the cluster remain analytic. The same Riesz-projection argument for $t \in \mathbb{R}^N$ appears in [5, Prop. 2.5] for a family with a fixed Dirichlet form and a multiplicative weight, which is an analogue rather than our variable-coefficient pencil. For the one-parameter theory see Kato [26, Ch. VII] (as quoted in [9, §4.1]).

3.2. **The first-order map is intrinsic and equivariant.** For an L^∞ symmetric 2-tensor $\dot{h}$ on B set $b_{\dot{h}}(e, e') := \frac{d}{ds}\big|_0 (q - \lambda_* m)_{h_0 + s\dot{h}}(e, e')$ for $e, e' \in E$.

Lemma 3.3. (a) *For every* $v \in T_{x_B}\mathcal{C}$ *there is a TT tensor* $\sigma(v)$*, depending linearly and only on* v*, such that* $A_{\mathrm{oct}}(v) := [b_v(e_i, e_j)] = A(\sigma(v)) := [b_{\sigma(v)}(e_i, e_j)]$*. Thus* A *is well defined on* T_B*.*
(b) $A(g \cdot \sigma) = \rho_E(g) A(\sigma) \rho_E(g)^{\mathsf{T}}$ *for all* $g \in G$*, of either orientation.*
(c) *For an octagon symmetry* γ *acting linearly on* $\mathcal{C}$*,* $\sigma(\gamma v) = \gamma_B \cdot \sigma(v)$*.*

Proof. Step 1 (first variation). Using $d(h^{ij}) = -h^{ik}\dot{h}_{kl}h^{lj}$ and $d\sqrt{h} = \frac{1}{2}(\operatorname{tr}\dot{h})\sqrt{h}$,

$$b_{\dot{h}}(e,e') = -\int_B \langle \dot{h}, T(e,e')\rangle\, dA, \qquad T(e,e') := \operatorname{Sym}(de\otimes de') - \tfrac{1}{2}\big(\langle de, de'\rangle - \lambda_* ee'\big)h_0.$$

The formula is tensorial and may be evaluated cellwise; it agrees with the b_v of Theorem 3.1, since the coefficients there are $a = \sqrt{h}\,h^{-1}$ and $\rho = \sqrt{h}$ and $t \mapsto h_t$ is C^1 into L^∞. (Compare [10, Lemma 2.1].)

Step 2 (T is divergence-free on E). Eigenfunctions are smooth and $\operatorname{tr}\nabla^2 e = -\lambda_* e$. A direct computation gives

$$2\operatorname{div} T = (\operatorname{tr}\nabla^2 e)\,de' + (\operatorname{tr}\nabla^2 e')\,de + \nabla^2 e(\nabla e', \cdot) + \nabla^2 e'(\nabla e, \cdot) - d\langle de, de'\rangle + \lambda_*(e\,de' + e'\,de) = 0.$$

Step 3 (Lie derivatives drop out). For a $W^{1,\infty}$ vector field X, $L_X h_0 = 2\operatorname{Sym}\nabla X^\flat \in L^\infty$, and integration by parts against the smooth T gives $b_{L_X h_0} = 2\int\langle X, \operatorname{div} T\rangle = 0$.

Step 4 (octagon tangents are TT plus a Lie derivative). Let $x(t)$ be a C^1 curve in $\mathcal{C}$ through x_B, $h_t = \Phi_t^* h_{B_t}$, and let $D_t\colon \mathbb{H}^2 \to \mathbb{H}^2$ be the developing map of h_t, which is ρ_t-equivariant, cellwise analytic in (t,p), and normalised by $D_0 = \mathrm{id}$. Fix a smooth radial kernel η_ε supported in an ε-ball and define the Riemannian centre of mass

$$\tilde{\Psi}_t(p) := \operatorname*{argmin}_{q\in\mathbb{H}^2} \int d(q, D_t(y))^2\, \eta_\varepsilon(d(p,y))\, dA(y).$$

The functional is strictly convex and proper, so $\tilde{\Psi}_t$ is well defined; it is ρ_t-equivariant because the kernel is deck-invariant; it is jointly C^∞ in (t,p) by the implicit function theorem (the Hessian in q is positive definite, p-derivatives fall on the kernel and t-derivatives on D_t inside the integral); $\tilde{\Psi}_0 = \mathrm{id}$; and $d\tilde{\Psi}_t = I + O(t)$, so $\tilde{\Psi}_t$ is a diffeomorphism for small $|t|$. It descends to $\Psi_t\colon B \to B_t$. Then $k_t := \Psi_t^* h_{B_t}$ is a smooth family of smooth hyperbolic metrics with $k_0 = h_0$, and $\iota_t := \Psi_t^{-1}\circ\Phi_t$ is an isometry $(B, h_t) \to (B, k_t)$ isotopic to the identity. The mixed derivative $\partial_t \nabla_p \iota_t$ exists and is continuous on each closed cell, so $X := \partial_t \iota_t|_0$ is Lipschitz and $\dot{h} = \dot{k} + L_X h_0$ almost everywhere.

By the Fischer–Marsden splitting, as stated in [29, Rem. 3.3], every symmetric tensor S decomposes orthogonally as $S = S_{tt} + L_X g + ((-\Delta_g f + f)g + g\nabla^2 f)$, where $S_{tt} + L_X g \in \ker dK_g$ and the last term lies in $(\ker dK_g)^\perp$. Applied to $S = \dot{k} \in \ker dK_{h_0}$, the last term vanishes, so $\dot{k} = \sigma + L_Y h_0$ with σ TT and Y smooth by elliptic regularity. Set $\sigma(v) := \sigma$. Steps 1 and 3 give $A_{\mathrm{oct}}(v) = A(\sigma(v))$. Two choices of Ψ_t differ by Diff_0, which changes $\dot{k}$ by a Lie derivative, so $\sigma(v)$ is independent of the choices; it is linear in v.

Step 5 (equivariance). Let g be an isometry of h_0, acting on TT tensors by $(g^{-1})^*$ and on E by $\rho_E(g)e = e\circ g^{-1}$. Then $T(e\circ g^{-1}, e'\circ g^{-1}) = (g^{-1})^* T(e,e')$ and dA is g-invariant, which gives (b). On $\operatorname{Re} Q(B)$ this is $\operatorname{Re}((g^{-1})^*\varphi)$ for holomorphic g and $\operatorname{Re}(\overline{(g^{-1})^*\varphi})$ for antiholomorphic g.

Relabelling. The fan mesh is equivariant under octagon symmetries, so $\Phi_{\gamma x} = \Gamma\circ\Phi_x\circ\gamma_B^{-1}$ and $h_{\gamma x(t)} = (\gamma_B^{-1})^* h_{x(t)}$. Since the TT projection is natural under isometries of h_0, this gives (c). □

4. Symmetry: characters and the K_0 reduction

4.1. **The algebraic model.** By [15, §7.1], $\mathrm{Isom}^+(B)/\langle r^4\rangle$ is the octahedral rotation group acting on $B/\langle J\rangle \cong \hat{\mathbb{C}}$ and permuting the six Weierstrass images; $\mathrm{Isom}^+(B) \cong GL(2,3)$ and $\mathrm{Isom}(B) \cong GL(2,3)\rtimes\mathbb{Z}/2$. Since a finite $S_4 \subset PSL(2,\mathbb{C})$ is conjugate to the standard octahedral group, whose unique orbit of size 6 is $\{0,\infty,\pm1,\pm i\}$, we may take $B \cong \{y^2 = x^5 - x\}$.

The maps

$$r\colon (x,y)\mapsto(ix,\zeta_8 y), \qquad t\colon x\mapsto\frac{x+i}{x-i}\ \text{(with } y\text{-factor } 2+2i), \qquad \kappa\colon (x,y)\mapsto(\bar{x},\bar{y})$$

generate a group of order 96, of which 48 elements are holomorphic and form a copy of $GL(2,3)$ (element orders 1:1, 2:13, 3:8, 4:6, 6:8, 8:12). This was verified in exact arithmetic over $\mathbb{Q}(\zeta_{24})$, including the curve identity for every element. Since $|\operatorname{Isom}(B)| = 96$ [15], this is all of G.

The space $Q(B) = \operatorname{Sym}^2 H^0(K_B)$ has basis $(dx/y)^2$, $x(dx/y)^2$, $x^2(dx/y)^2$. An automorphism with Möbius matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ and y-factor μ pulls back $(dx/y,\, x\,dx/y)$ by $P = (\det/\mu)\left(\begin{smallmatrix} d & b \\ c & a \end{smallmatrix}\right)$; for holomorphic g the real character on $T_B \cong Q(B) \cong \mathbb{R}^6$ is $\chi_T(g) = 2\operatorname{Re}\operatorname{tr}\operatorname{Sym}^2 P$. For orientation-reversing g the action is $\mathbb{C}$-antilinear, and an antilinear map has real trace 0.

Proposition 4.1 (Characters; exact)**.** *On T_B: $\chi_T(r) = \chi_T(r^3) = \chi_T(r^2) = -2$, $\chi_T(J) = 6$, $\chi_T(t) = 0$, and $\chi_T(g) = 0$ for every orientation-reversing g. Hence $\langle \chi_T, \chi_T \rangle = 2$ and $\langle \chi_T, 1 \rangle = 0$; in particular $T_B^G = 0$.*

These values were also reproduced with an independently written pull-back model.

4.2. **The antipodal splitting.** The antipodal isometry a has conjugacy class $\{a, a^3 = aJ\}$ of size 2, so a is central modulo J, and $a^2 = J$. Since $\chi_T(J) = 6 = \dim T_B$, J acts trivially on T_B, so $a^2 = \mathrm{id}$ on T_B and

$$T_B = W_0 \oplus K_0, \qquad W_0 := \{a = +1\}, \quad K_0 := \{a = -1\},$$

both G-invariant. Exactly, both have dimension 3, $\langle \chi_\pm, \chi_\pm \rangle = 1$ and $\langle \chi_+, \chi_- \rangle = 0$. Hence K_0 and W_0 are absolutely irreducible and non-isomorphic. (In the labelling of Cook's character table [7, App. A], $K_0 \cong \chi_7$ and $W_0 \cong \chi_{10}$; this is not used.) On the octagon elements, $\chi_{K_0}(r) = -1$, $\chi_{K_0}(s_{\mathrm{mid}}) = 1$, $\chi_{K_0}(s_{\mathrm{vert}}) = -1$, and $\chi_{W_0}(r) = -1$, $\chi_{W_0}(s_{\mathrm{mid}}) = -1$, $\chi_{W_0}(s_{\mathrm{vert}}) = 1$; both vanish on t.

Lemma 4.2 (Isolation of E; see Lemma 7.2)**.** $\dim E = 3$ *and E is an irreducible real G-module.*

Proposition 4.3. $K_0 \subset \ker A$.

Proof. E is irreducible of odd real dimension, so $\operatorname{End}_G(E) = \mathbb{R}$ and the central element J acts on E by ± 1. If $J = -1$ then $\rho_E(a)^2 = -I$ on $\mathbb{R}^3$, which is impossible ($\det^2 = -1$). So J acts trivially, $\rho_E(a)$ commutes with $\rho_E(G)$ (as a is central modulo J), and $\rho_E(a) = \pm I$. By Lemma 3.3(b), $A(a \cdot v) = A(v)$. For $v \in K_0$, $a \cdot v = -v$, so $A(v) = 0$. □

Proposition 4.4. *If $A \neq 0$, then* $\ker A = K_0$ *and* $\operatorname{rank} A = 3$.

Proof. $A|_{W_0} \colon W_0 \to \operatorname{Sym}(E)$ is G-equivariant and W_0 is irreducible, so $A|_{W_0}$ is 0 or injective. Since $A \neq 0$ and $A|_{K_0} = 0$, it is injective. □

Proposition 4.5. *Every G-invariant symmetric bilinear form on K_0 is $c \cdot g_{\mathrm{inv}}$ for a fixed G-invariant inner product g_{inv} and some $c \in \mathbb{R}$.*

Proof. K_0 is absolutely irreducible (Schur). □

Remark 4.6 (Consistency, not used). The full Clifford–Gallagher computation shows $\operatorname{Sym}^2_0 E \cong D \oplus W_0$ with D two-dimensional irreducible, for each of the four candidate three-dimensional irreducibles. A certified computation gives $\operatorname{tr}(g|E) \in [1 \pm 0.0352]$ for $g = r, s_{\mathrm{mid}}, s_{\mathrm{vert}}$ at $n = 96$, so E is the octahedral representation, in agreement with [7, Thm. 3.9]. Neither fact is used.

4.3. **Why the octagon group is not enough.** Restricted to the octagon group D_8 (16 elements), K_0 has norm $(9{+}9{+}4{\cdot}1{+}2{\cdot}1{+}4{\cdot}1{+}4{\cdot}1)/16 = 2$, so $K_0|_{D_8} = \varepsilon_s \oplus (2\text{-dim})$. D_8-invariant quadratic forms on K_0 thus form a two-parameter family, and one certified direction controls only the ε_s-line. The non-octagon element t is genuinely needed, both for irreducibility of W_0 and for Proposition 4.5; the corresponding invariance of the Hessian is proved in Lemma 6.4.

4.4. **The equiangular family.** Let $F_B(p)$ be the octagon with all vertices at hyperbolic radius $R(p)$, V_{2j} at angle $j\pi/2$ and V_{2j+1} at angle $\varphi + j\pi/2$, where $p = \varphi - \pi/4$. The $\pi/2$-rotation and the reflection $\theta \mapsto \varphi - \theta$ act transitively on the vertices, so all angles are equal, and $L_{k+4} = L_k$ by the π-rotation. Splitting the angle at V_0 along the spokes, the surface condition (angle $= \pi/4$) is

$$C^2xy - C(x+y) - 1 = 0, \qquad C = \cosh R,\ x = \tan(\varphi/2),\ y = \tan(\pi/4 - \varphi/2),$$

so that $C = 3 + 2\sqrt{2}$ at $p = 0$. (The Taylor jets of the constraints along this curve were certified to vanish to second order in Arb.) $F_B(p)$ is invariant under r^2 and s_{mid}, and r maps $F_B(p)$ to $F_B(-p)$. Write $x(p) = F_B(p)$, $v_B = x'(0)$ and $w_B := \sigma(v_B)$.

Lemma 4.7. *$w_B \in K_0 \setminus \{0\}$.*

Proof. By exact linear algebra, $\{v \in T_B : rv = -v,\ s_{\mathrm{mid}}v = v\}$ is a line, and its spanning vector satisfies $av = -v$, so the line lies in K_0. By Lemma 3.3(c), $rw_B = -w_B$ and $s_{\mathrm{mid}}w_B = w_B$, so w_B lies on this line.

To see $w_B \neq 0$: the pairing g_0 of $F_B(p)$ is the translation along the bisector of side 0, and hyperbolic trigonometry gives $d\ell(g_0)/dp = -\cos(\pi/4)/(\sin^2(\pi/8)\sinh r_{\mathrm{in}}) = -\sqrt{2+2\sqrt{2}} = -2.19736822693562$ and $d\ell(g_1)/dp = +2.19736822693562$, where r_{in} is the inradius of the regular octagon. Thus $\ell(g_0) - \ell(g_1)$, a function on $\mathcal{T}_2$, has nonzero derivative along $\pi \circ F_B$. By Lemma 6.3(b) below, $d(\tau \circ x)(v_B) = d\tau(w_B)$ for length-type functions τ, so $w_B \neq 0$. (The certificate of Proposition 6.8 also forces $w_B \neq 0$.) □

Remark 4.8. The reflection type matters: for s_{vert}-type reflections the line $\{rv = -v,\ sv = v\}$ lies in W_0, not in K_0. So the identification of the symmetry of F_B as an s_{mid}-type reflection (fixing four Weierstrass points) is load-bearing. Here s_{mid} fixes the centre, the vertex point, the midpoint class on its axis and one further midpoint class via the gluing; s_{vert} fixes only the centre and the vertex point.

5. First-order analysis

Proposition 5.1. *$\operatorname{tr} A \equiv 0$; equivalently $d\Lambda(x_B) = 0$ on $T_{x_B}\mathcal{C}$.*

First proof (octagon coordinates). The map $r\colon \mathbb{R}^{16} \to \mathbb{R}^{16}$ is linear, maps $\mathcal{C}$ to $\mathcal{C}$, and B_{rx} is isometric to B_x, so $\Lambda \circ r = \Lambda$. Hence $d\Lambda = d\Lambda \circ P$ on $T := T_{x_B}\mathcal{C}$, with $P = \frac{1}{8}\sum_{j<8} r^j$ the projection onto T^r. The fixed space $(\mathbb{R}^{16})^r = \{\delta_k = \mathrm{Rot}^k\delta_0\}$ is two-dimensional (radial and tangential). The radial vector is not in T, because $d(\text{angle sum})/dR = -5.5729711960 \neq 0$ (certified; Proposition 7.6; it equals the closed form from $\cot(\theta/2) = \cosh R \tan(\pi/8)$). The tangential vector is the infinitesimal rotation, along which Λ is constant. So T^r is spanned by the rotation and $d\Lambda = 0$. By Theorem 3.1(4), $d\Lambda(v) = \operatorname{tr} A(v)$. □

Second proof (Schur). $\operatorname{tr} A$ is a G-invariant linear functional on T_B (Lemma 3.3(b)), and $\langle \chi_T, 1\rangle = 0$ (Proposition 4.1). □

In fact G-equivariance under the single element r suffices, since r has no fixed vector on T_B; this is how the statement is formalised in Section 9.

Proposition 5.2 ($A \neq 0$; certified)**.** *Let $\delta \in T_{x_B}\mathcal{C}$ be the alternating-radius tangent: vertex radii $R + t(-1)^k + \varsigma(t)$, angles $k\pi/4$, with $\varsigma(t)$ restoring the angle sum and $\varsigma'(0) = 0$. Then there is an exact eigenfunction $e \in E$ with*

$$b_\delta(e,e) \leq -3.0383476242423058 \cdot m(u) < 0$$

($n = 192$; at $n = 96$ the bound is $-0.9989201873540878 \cdot m(u)$; under Hypothesis 1.1 the bounds are $-3.0875694043934345 \cdot m(u)$ and $-1.0282409843057394 \cdot m(u)$), where u is the finite-element function of Section 7.5 and $e = P_E u$. Consequently $A(\sigma(\delta)) \neq 0$.

The deformation δ preserves $L_{k+4} = L_k$ by the π-rotation symmetry; ς is analytic by the implicit function theorem since $\partial(\sum \theta)/\partial\varsigma = -5.573 \neq 0$, and $\varsigma'(0) = 0$ because all $\partial \sum \theta/\partial R_k$ are equal at x_B while $\sum(-1)^k = 0$. Up to scale δ is the tangent of the equilateral family F_A. The proof is in Section 7.5. Identifying $[b_\delta(e_i, e_j)]$ with the first-order variation along δ only needs the one-parameter case of Theorem 3.1.

Corollary 5.3. *$K := \ker A = K_0$, and* $\operatorname{rank} A = 3$.

Proof. Propositions 4.3, 4.4 and 5.2. □

6. Second-order analysis

6.1. A local-maximum criterion for a degenerate cluster.

Lemma 6.1. *Let $M(v)$ be a symmetric 3×3 matrix function on a neighbourhood of $0 \in \mathbb{R}^6$ with $M(v) = \lambda_* I + A(v) + Q_2(v) + R(v)$, where A is linear, Q_2 is quadratic and $|R(v)| = o(|v|^2)$. Let $K = \ker A$. Assume*

(a) $\operatorname{tr} A \equiv 0$;
(b) $\operatorname{tr} Q_2(k) \le -\kappa|k|^2$ *for all $k \in K$, for some $\kappa > 0$.*

Then $\lambda_{\min} M(v) < \lambda_*$ for all sufficiently small $v \neq 0$.

Proof. Let W be a complement of K; $A|_W$ is injective. Each $A(w)$, $w \neq 0$, is traceless and nonzero and so has a negative eigenvalue; by compactness $c_A := \min_{w\in W, |w|=1}(-\lambda_{\min} A(w)) > 0$. This is the first use of (a). Write $v = w + k$ with $w \in W$, $k \in K$, and choose C'' with $\|Q_2(v)\| + |R(v)| \le C''(|w|^2 + |k|^2)$ on a small ball.

Case (i): $|w| \ge C|k|^2$, with $C > 4C''/c_A$. Then $w \neq 0$, and by Weyl's inequality

$$\lambda_{\min} M(v) - \lambda_* \le -c_A|w| + C''|w|^2 + C''|k|^2 \le -c_A|w| + C''|w|^2 + \tfrac{c_A}{4}|w| < -\tfrac{c_A}{2}|w| + C''|w|^2,$$

which is negative for small v.

Case (ii): $|w| < C|k|^2$. Then $k \neq 0$ and $|v| \asymp |k|$. Using $\operatorname{tr} A(w) = 0$ (second, essential use of (a)) and $\operatorname{tr} Q_2(w + k) = \operatorname{tr} Q_2(k) + O(|k|^3)$,

$$\operatorname{tr}\,(M(v) - \lambda_* I) = \operatorname{tr} Q_2(k) + O(|k|^3) + o(|k|^2) \le -\kappa|k|^2 + o(|k|^2) < 0,$$

so $\lambda_{\min} M(v) \le \lambda_* + \frac{1}{3}\operatorname{tr}(M(v) - \lambda_* I) < \lambda_*$. □

The criterion is sufficient, not necessary; the sharper condition is $\lambda_{\min} Q_2(k) < 0$ on K. Note also that Λ itself need not have a local maximum: off K it may increase at second order. Since $\Lambda = \operatorname{tr} M$, we have $\operatorname{tr} Q_2(k) = \frac{1}{2} D^2\Lambda(k, k)$.

6.2. The Teichmüller chart.

For a word w in the generators, let

$$\tau_w(x) := \operatorname{tr}_{SO(2,1)} w(g_0(x), \dots, g_3(x)).$$

We use the six words

$$g_0 g_1, \quad g_0 G_2, \quad g_1 g_2, \quad g_1 g_3, \quad g_1 G_3, \quad g_2 g_3 \qquad (G_k = g_k^{-1}).$$

Proposition 6.2 (Certified; Section 7.7)**.** *The 6×11 Jacobian of $(\tau_w)_w$ on $T_{x_B}\mathcal{C}$ has rank 6.*

Lemma 6.3. *Let $\mathcal{S} \subset \mathcal{C}$ be the analytic 6-dimensional coordinate slice through x_B in the free coordinates $\{3, 6, 10, 12, 13, 15\}$ (Section 7.7). Then:*

(a) $\pi|_{\mathcal{S}}$ *is a homeomorphism from a neighbourhood of* x_B *in* $\mathcal{S}$ *onto an open neighbourhood* N *of* B *in* $\mathcal{T}_2$*, and the functions* $\bar\tau_w$ *with* $\tau_w = \bar\tau_w \circ \pi$ *are local coordinates of* $\mathcal{T}_2$ *at* B*.*
(b) *For every word* w *and every* C^1 *curve* $x(t)$ *in* $\mathcal{C}$ *through* x_B*,* $d(\tau_w \circ x)(v) = d\tau_w(\sigma(v))$*, where* $d\tau_w$ *is the derivative of the* Diff_0*-invariant length-type function on smooth metrics. In particular* $(d\tau_w)_w$ *is injective on TT tensors, and* $\sigma\colon T_{x_B}\mathcal{S} \to T_B$ *is an isomorphism.*
(c) *There is an analytic retraction* $p\colon \mathcal{C} \to \mathcal{S}$ *near* x_B *with* $\pi \circ p = \pi$*. Hence* $\Lambda_{\mathcal{C}} = \Lambda_{\mathcal{S}} \circ p$*,* $d\Lambda_{\mathcal{S}}(x_B) = 0$*,* $D^2\Lambda_{\mathcal{C}}(w,w) = D^2\Lambda_{\mathcal{S}}(dp\,w, dp\,w)$*, and* $\sigma \circ dp = \sigma$ *on* $T_{x_B}\mathcal{C}$*.*
(d) *Theorem 3.1 applies on* $\mathcal{S}$*:* $M_{\mathcal{S}}(v) = \lambda_* I + A(\sigma v) + Q_2(v) + O(|v|^3)$ *and* $\operatorname{tr}(A \circ \sigma) \equiv 0$*.*

Proof. (a) $\mathrm{tr}_{SO(2,1)}$ is conjugation-invariant and needs no sign choice, so $\tau_w = \bar\tau_w \circ \pi$. $\Phi := (\tau_w)_w$ is analytic on $\mathcal{C}$ and its derivative on $T_{x_B}\mathcal{S}$ has rank 6, so $\Phi|_{\mathcal{S}}$ is an analytic diffeomorphism near x_B onto an open $U \subset \mathbb{R}^6$. Hence $\pi|_{\mathcal{S}}$ is continuous and injective. $\mathcal{T}_2$ is a topological 6-manifold (Fenchel–Nielsen coordinates; [13, §2]), so by invariance of domain $\pi|_{\mathcal{S}}$ is a homeomorphism onto an open $N \ni B$.

(b) With k_t the smooth comparison family of Lemma 3.3, Step 4, $\tau_w(x(t)) = \tau_w(k_t)$, and $\dot k = \sigma(v) + L_Y h_0$. The function $\tau_w(k)$ is the trace of the holonomy along a fixed loop of the $\mathfrak{so}(2,1)$-valued Cartan connection $\omega_{LC} + \theta$ of k. This is defined and C^1 in k for every metric, and it is flat, hence invariant under free homotopy, exactly when the curvature is -1. So τ_w is a Diff_0-invariant C^1 function on hyperbolic metrics, and the Lie-derivative term drops out. Rank 6 then forces σ to be onto the 6-dimensional TT space, and injectivity of $(d\tau_w)_w$ on TT follows.

(c) Set $p := (\Phi|_{\mathcal{S}})^{-1} \circ \Phi$. Since $\bar\tau$ is injective on N, $\pi \circ p = \pi$. As $dp|_{T\mathcal{S}} = \mathrm{id}$ and $d\Lambda_{\mathcal{C}}(x_B) = 0$, the Hessians correspond. Applying the injective $(d\tau_w)_w$ to $d\tau(\sigma(dp\,v)) = d(\tau \circ p)(v) = d\tau(\sigma(v))$ gives $\sigma \circ dp = \sigma$.

(d) $\mathcal{S}$ is an analytic sub-family of $\mathcal{C}$; use Lemma 3.3(a) and Proposition 5.1. □

6.3. **Invariance of the Hessian under all of G.**

Lemma 6.4. *The quadratic form* $H(\varsigma) := D^2\Lambda_{\mathcal{S}}(\sigma^{-1}\varsigma, \sigma^{-1}\varsigma)$ *on TT tensors is invariant under the natural action of* G*. Consequently* $H|_{K_0} = c \cdot g_{\mathrm{inv}}$ *for some* $c \in \mathbb{R}$*.*

Proof. Let $g \in G$ act on $\mathcal{T}_2$ by change of marking, normalised by $g \cdot [k] = [(g^{-1})^* k]$. Then g fixes B, and $\Lambda \circ g = \Lambda$ since the surfaces are isometric.

Step 1 (g is analytic in the chart). For every word w', $\bar\tau_{w'}(g \cdot X) = \bar\tau_{g^{-1}(w')}(X)$, because the $(g^{-1})^*k$-length of the class w' is the k-length of $g^{-1}(w')$. (For orientation-reversing g, $g^{-1}(w')$ is defined up to conjugation and inversion, which does not affect $\mathrm{tr}_{SO(2,1)}$.) Since each $\tau_{w''}|_{\mathcal{S}}$ is analytic and the chart is $(\Phi|_{\mathcal{S}})^{-1} \circ \bar\tau$, the map $g_{\mathcal{S}} := (\pi|_{\mathcal{S}})^{-1} \circ g \circ \pi|_{\mathcal{S}}$ is analytic near x_B and fixes x_B. This uses continuity of the extended mapping class group action on $\mathcal{T}_2$, so that $g \cdot \pi(x) \in N$ for x near x_B. As $d\Lambda_{\mathcal{S}}(x_B) = 0$, the Hessian is invariant: $D^2\Lambda_{\mathcal{S}}(dg_{\mathcal{S}}v, dg_{\mathcal{S}}v) = D^2\Lambda_{\mathcal{S}}(v,v)$.

Step 2 (σ intertwines). Let $x(t)$ be a curve in $\mathcal{S}$ and $y = g_{\mathcal{S}} \circ x$. By Lemma 6.3(b), $d\tau_w(\sigma(y')) = \frac{d}{dt}\tau_{g^{-1}w}(x(t)) = d\tau_{g^{-1}w}(\sigma(x'))$. On hyperbolic metrics $\tau_{g^{-1}w}(k) = \tau_w((g^{-1})^*k)$; differentiating at the g-fixed h_0 gives $d\tau_{g^{-1}w}(\varsigma) = d\tau_w(g \cdot \varsigma)$, since $(g^{-1})^* L_Y h_0 = L_{g_*Y} h_0$ is killed by Diff_0-invariance. Injectivity of $(d\tau_w)_w$ on TT gives $\sigma(y') = g \cdot \sigma(x')$.

Steps 1 and 2 give $H(g \cdot \varsigma) = H(\varsigma)$ for all $g \in G$, in particular for t. Proposition 4.5 gives the last claim. □

Remark 6.5. An alternative route would cite real-analyticity of π into the standard analytic structure of $\mathcal{T}_2$ and analyticity of the action of the stabiliser of B in the extended mapping class group with derivative equal to the natural action on $\mathrm{Re}\, Q(B)$. The argument above avoids these inputs by working entirely in the certified chart $\mathcal{S}$.

6.4. **The Ky Fan bound along F_B.**

Lemma 6.6 (Ky Fan trace principle)**.** *For every p and every 4-dimensional $V \subset H^1(B)$, $\Lambda(p) \le \operatorname{tr}_V(q_p \,|\, m_p)$, with equality when V is the span of 1 and the cluster eigenfunctions.*

Proof. Let $\mu_0 \le \cdots \le \mu_3$ be the eigenvalues of $q_p|_V$ relative to m_p. For each j, the span of the first $j+1$ of their eigenvectors is a $(j+1)$-dimensional subspace on which the Rayleigh quotient is at most μ_j, so by min-max ([7, Thm. 1.22], from Chavel) $\lambda_j \le \mu_j$. Summing and using $\lambda_0 = 0$ gives the claim. (For the finite-dimensional statement see [30], attributing it to [11, 12].) □

Take $V(p) = \operatorname{span}\{1, e_i + p\psi_i\}$, where $e_i \in E$ are exact eigenfunctions and ψ_i are fixed finite-element correctors, and set $g(p) = \operatorname{tr}(G_m(p)^{-1} G_q(p))$ with the Gram matrices of m_p, q_p on $V(p)$. Then $g \ge \Lambda \circ x$ and $g(0) = \Lambda(0)$. Both are analytic, so $g - \Lambda \circ x \ge 0$ has a minimum at 0 and

$$(\Lambda \circ x)''(0) \le g''(0). \tag{2}$$

Even without twice differentiability, the lim sup of the second difference quotients of $\Lambda \circ x$ is at most $g''(0)$; the analyticity of Theorem 3.1 is needed only to read the result as a statement about $D^2\Lambda$.

Lemma 6.7. *With $N = M_0^{-1}$,*

$$g''(0) = \operatorname{tr}(NX) - 2\operatorname{tr}(NM_1NA) + 2\lambda_*\, c_1^{\mathsf{T}} N c_1 / m(1,1),$$

where $X_{ij} = (q'' - \lambda_* m'')(e_i, e_j) + 2b(\psi_i, e_j) + 2b(e_i, \psi_j) + 2(q - \lambda_* m)(\psi_i, \psi_j)$, $A_{ij} = b(e_i, e_j)$ with $b = q' - \lambda_* m'$, $M_{1,ij} = m'(e_i, e_j) + m(\psi_i, e_j) + m(e_i, \psi_j)$, $c_{1,i} = m'(e_i, 1) + m(\psi_i, 1)$ and $M_{0,ij} = m(e_i, e_j)$.

Proof. Expand $G_m^{-1} G_q$ to second order in p, using the Schur complement of the constant function, $q_0(\psi, e) = \lambda_* m_0(\psi, e)$, $Q(0) = \lambda_* M_0$ and $m(e, 1) = 0$. The basis e_i need not be orthonormal. (The formula was re-derived independently and checked against an exact pencil test to relative accuracy 3×10^{-39}, and against a five-point finite difference of g.) □

Proposition 6.8 (Certified; Section 7.6)**.** *Along F_B, $g''(0) \le -9.341154560446736$ ($n = 96$, P_2 elements), with central value $[-14.274 \pm 2.13 \times 10^{-4}]$. Consequently*

$$D^2\Lambda_{\mathcal{C}}(v_B, v_B) = (\Lambda \circ x)''(0) \le -9.3411 < 0.$$

Under Hypothesis 1.1 the same certificate gives $g''(0) \le -12.850476214196528$.

The last equality holds because $(\Lambda \circ x)''(0) = D^2\Lambda_{\mathcal{C}}(v_B, v_B) + d\Lambda_{\mathcal{C}}(x''(0))$ and $d\Lambda_{\mathcal{C}}(x_B) = 0$ (Proposition 5.1).

7. Certified computations

All computations below use Arb ball arithmetic through python-flint 0.9.0 (96- or 128-bit precision), exact rationals or exact arithmetic in $\mathbb{Q}(\zeta_{24})$ (with sympy 1.14.0 for Sturm sequences). Floating-point computations (numpy/scipy) only propose candidates. Every truncation carries an explicit bound.

7.1. **Isolation of the cluster by the trace formula.** We use the Selberg trace formula in the normalisation of [15, (3.1)], which follows Buser [4, p. 253]: for every admissible f and closed hyperbolic surface S of genus g,

$$\sum_{\lambda \in \sigma(S)} \hat f(r(\lambda)) = 2(g-1) \int_0^\infty r \hat f(r) \tanh(\pi r)\, dr + (2\pi)^{-1/2} \sum_\gamma \frac{\Lambda(\gamma) f(\ell(\gamma))}{2\sinh(\ell(\gamma)/2)},$$

with the conventions of [15]. We use [15, Lemma 3.2]: if $0 < a \le \lambda_1(S)$, $a < b$, $\hat f(r(\lambda)) \ge c > 0$ on $[a, b]$, $\hat f(r(\lambda)) \ge 0$ for $\lambda \ge b$, and $f(x) \le 0$ for $x \ge \operatorname{sys}(S)$, then the number of eigenvalues in

$[a,b]$ is at most $c^{-1}(I+G_P-\hat{f}(i/2))$, where I is the integral term and G_P the geodesic sum over any set P of unoriented primitive closed geodesics. We also use the following variant, whose proof is that of [15, Lemma 3.2] unchanged.

Lemma 7.1. *Let $F(\lambda) := \hat{f}(r(\lambda))$. If $F \geq c > 0$ on $[0,a]$, $F \geq 0$ on $[a,\infty)$ and $f \leq 0$ on $[\mathrm{sys}(S),\infty)$, then $\#(\sigma(S)\cap(0,a]) \leq (I+G_P-F(0))/c$.*

Test functions. $f(x) = u(x^2/s^2)e^{-x^2/(2s^2)}$ and $\hat{f}(\xi) = s\,v(s^2\xi^2)e^{-s^2\xi^2/2}$, with $u(y) = \sum_{n\leq N} c_n H_{2n}(\sqrt{y})$, $v(y) = \sum_{n\leq N}(-1)^n c_n H_{2n}(\sqrt{y})$ (H the physicists' Hermite polynomials), $c_n, s \in \mathbb{Q}$ and $N = 21$. Since $r(\lambda)^2 = \lambda - \frac{1}{4}$, $F(\lambda) = s\,v(s^2(\lambda-\frac{1}{4}))e^{-s^2(\lambda-1/4)/2}$. ($N$ must be odd: otherwise the leading coefficients $c_N 4^N$ and $(-1)^N c_N 4^N$ make $u \leq 0$ and $v \geq 0$ near $+\infty$ incompatible.)

Verification. The sign conditions $u \leq 0$ on $[y_0,\infty)$ (with rational $y_0 \leq \ell_1^2/s^2$) and $v \geq 0$ beyond the window are proved by exact Sturm counts: 0 roots in each case. A lower bound c for F on the window comes from 4000 Arb balls. The integral term uses exact Gaussian moments $\int_0^\infty r(s^2r^2)^k e^{-s^2r^2/2}dr = 2^k k!/s^2$ plus a rigorous `acb.integral` on $[0,12]$ with an explicit tail bound. The geodesic term uses the lower bounds $12, 12, 24$ on the unoriented multiplicities of $\ell_1 = 2\,\mathrm{arccosh}(1+\sqrt{2})$ (the systole), $\ell_2 = 2\,\mathrm{arccosh}(3+2\sqrt{2})$ and $\ell_3 = 2\,\mathrm{arccosh}(5+3\sqrt{2})$ [15, §7.4]; these are primitive since $\ell_3 < 2\ell_1$, and only lower bounds are needed because $f \leq 0$ there.

Lemma 7.2 (Isolation and multiplicity)**.** (a) $\mathrm{Spec}(B)\cap(0,3.3] = \emptyset$ *and* $\#(\mathrm{Spec}(B)\cap[3.3,4.0]) \leq 3$*, with multiplicity.*

(b) $\lambda_3(B) \leq 3.8391848048$*. Hence* $\lambda_1,\lambda_2,\lambda_3 \in (3.3, 3.8391849]$ *and* $\lambda_4 > 4.0$*, a gap of at least* 0.1608*. This part uses neither Hypothesis 1.1 nor the multiplicity theorem of* [15].

(c) $\lambda_1 = \lambda_2 = \lambda_3 \leq 3.8388872830$*,* $\dim E = 3$*, and E is an irreducible real G-module.*

(d) $\lambda_1 \geq 3.8388757010$ *(Proposition 7.4). Hence* $\lambda_1 \in [3.8388757010, 3.8388872830]$.

Proof. (a) Run A applies Lemma 7.1 with $N = 21$, $s = 1$ on $[0, 3.3]$: $c \geq 0.9816424759937855$ and the certified bound is $[0.912612352675290375 \pm 9.95\times10^{-20}] < 1$. Run B applies [15, Lemma 3.2] with $a = 3.3 \leq \lambda_1$ (from run A), $N = 21$, $s = 13/10$ on $[3.3, 4.0]$: $c \geq 0.9964283521820437$ and the bound is $[3.86966111976973759 \pm 1.65\times10^{-18}] < 4$.

(b) Section 7.2.

(c) By [15, Lemmas 7.2–7.3] with $j = 1$, whose proofs use symmetry, nodal domains and [15, Prop. 4.4] but no eigenvalue enclosure, the λ_1-eigenspace contains no 1- or 2-dimensional irreducible real representation of G. So $m_1 \geq 3$, and with (b), $\lambda_1 = \lambda_2 = \lambda_3$ and $\dim E = 3$; any decomposition of E would contain a summand of dimension 1 or 2, so E is irreducible. The certified P_2 Rayleigh–Ritz bound $\lambda_3 \leq 3.8388872829261946$ (Section 7.2) gives the upper end.

(d) Section 7.3; it uses only (a).

Cross-check (not used). By (a) and (b) the only eigenvalue in $(0, 4.0]$ is λ_1, so the eigenvalue of Hypothesis 1.1, if assumed, is λ_1; its interval is consistent with (d). □

As sanity checks, windows that contain known eigenvalues give bounds above the true counts: $3.4803 \geq 3$ for $[3.3, 3.9]$ and $7.5466 \geq 7$ for $[3.3, 5.4]$. The window end $X = 4.05$ fails (bound 4.0536); with only three geodesic lengths the method does not reach beyond about 4.05.

7.2. **Conforming finite elements on the glued octagon.** P_1 *elements.* On each geodesic triangle of the mesh, trial functions are affine in the barycentric parameters (a,b) of Section 2.4. Isometries act linearly on W, so the construction is chart-free, and the trace on an edge depends only on the two endpoint values. The space is therefore *exactly conforming on the glued surface*, including across the projective side pairings, it lies in $H^1(B)$, and it contains the constants. (Klein-coordinate P_1 elements lack this property.)

P_2 elements. Polynomials of degree ≤ 2 in (a, b). On an edge, $W = sP_i + (1-s)P_j$ and the trace is a quadratic in s determined by the two endpoint values and the value at $s = \frac{1}{2}$, the geodesic midpoint. Isometries preserve s, so this space is also exactly conforming and contains the constants. Edge degrees of freedom are keyed by glued endpoint labels; the code asserts Euler characteristic -2 and exactly two triangles per edge.

Element integrals. All entries are integrals $\int_{\text{simplex}} p \cdot Q^{-s}$ with $s \in \{\frac{1}{2}, \frac{3}{2}\}$ (P_1) or $s \in \{\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \frac{7}{2}\}$ (P_2). Write $Q = Q_c(1+\epsilon)$ and expand $(1+\epsilon)^{-s}$ in a truncated binomial series (4 terms for P_1, 6 for P_2) with a Lagrange remainder, e.g. for P_1 $|R| \le |\binom{-s}{4}| \epsilon_{\max}^4 (1-\epsilon_{\max})^{-s-4}$; monomials are integrated exactly by $\int a^i b^j = i!\, j!/(i+j+2)!$.

Rayleigh–Ritz. Untrusted float eigenvectors x_1, x_2, x_3 and the constant span a 4-dimensional V; its 4×4 matrices K, M are summed exactly in Arb. By min-max [7, Thm. 1.22], $\lambda_3(B) \le \mu_3(K, M)$, and $\mu_3 < t$ is certified by Sylvester's criterion for $tM - K$, with t found by bisection. Applied to the same space, Lemma 6.6 gives a certified upper bound for $\lambda_1 + \lambda_2 + \lambda_3$.

The certified bounds are in Tables 3 and 4. At $n = 192$ (P_1), $\lambda_3(B) \le 3.8391848048$; at $n = 96$ (P_2), $\lambda_3(B) \le 3.8388872830$ and $\lambda_1 + \lambda_2 + \lambda_3 \le 11.516661838577997$. The float P_2 errors decrease like h^4 (4.8×10^{-6}, 3.0×10^{-7}, 1.9×10^{-8}). The discrete cluster splits as $2 + 1$ because the mesh has only the octagon's symmetry.

7.3. **A guaranteed lower bound for λ_1: Lehmann–Goerisch.** Rayleigh–Ritz gives only upper bounds. For the lower bound we use the Lehmann–Goerisch method in the form of [28, §4.1, Thms. 4.1 and 4.3], which goes back to Goerisch; the same statement, written with an explicit shift, is [1, Thm. 2.1] as reproduced in [37, §2]. We quote it for our setting. Let D be a real vector space with symmetric bilinear forms M (positive definite) and N whose eigenvalue problem $M(u, \varphi) = \tilde{\lambda} N(u, \varphi)$ has a discrete spectrum $\tilde{\lambda}_1 \le \tilde{\lambda}_2 \le \cdots$ with an M-orthonormal eigenbasis. Let X be a real vector space with a positive semidefinite symmetric form b_G and $T\colon D \to X$ linear with $b_G(Tu, Tv) = M(u, v)$. Given $v_1, \dots, v_n \in D$, let $w_1, \dots, w_n \in X$ satisfy $b_G(w_i, T\varphi) = N(v_i, \varphi)$ for all $\varphi \in D$. For $\rho > 0$ put $A_0 = (M(v_i, v_j))$, $A_1 = (N(v_i, v_j))$, $A_2 = (b_G(w_i, w_j))$, $A = A_0 - \rho A_1$ and $B = A_0 - 2\rho A_1 + \rho^2 A_2$, and suppose B is positive definite; let $\nu_1 \le \cdots \le \nu_n$ be the eigenvalues of $Az = \nu Bz$ and q the number of negative ones.

Theorem 7.3 (Lehmann–Goerisch; quoted from [28])**.** *If $\rho \le \tilde{\lambda}_{m+1}$ for some $m \ge n$, then $\tilde{\lambda}_{m+1-k} \ge \rho - \rho/(1-\nu_k)$ for $1 \le k \le q$.*

Instantiation. Take $D = H^1(B)$, $M = q + m$ and $N = m$. The eigenvalues are shifted, $\tilde{\lambda} = \lambda + 1$: $\tilde{\lambda}_1 = 1$ (the constants), $\tilde{\lambda}_{2,3,4} = \lambda_{1,2,3} + 1$ and $\tilde{\lambda}_5 = \lambda_4 + 1$. By Lemma 7.2(a), $\lambda_4 > 4.0$, so $\rho := 5 \le \tilde{\lambda}_5$; we take $m = 4$ and $n = 3$. Nothing else about the spectrum is used. In particular neither Hypothesis 1.1 nor [15, Lemmas 7.2–7.3] enters. Let $X = L^2(TB) \times L^2(B)$, $Tu = (\nabla u, u)$, and let b_G be the L^2 product, so $b_G(Tu, Tv) = M(u, v)$. For trial functions v_i and any $\sigma_i \in H(\operatorname{div}, B)$ put $w_i = (\sigma_i,\, v_i + \operatorname{div} \sigma_i)$. Integration by parts on the closed surface gives, for all $\varphi \in H^1(B)$,

$$b_G(w_i, T\varphi) = \int \sigma_i \cdot \nabla\varphi + \int (v_i + \operatorname{div}\sigma_i)\varphi = \int v_i \varphi = N(v_i, \varphi),$$

exactly and with no constraint on $\operatorname{div}\sigma_i$. This is why the shift is used. The choice of σ_i affects only the sharpness. With $k = 3$, $m + 1 - k = 2$, so the theorem bounds $\tilde{\lambda}_2 = \lambda_1 + 1$, the first *nonzero* eigenvalue. Since $\lambda_2, \lambda_3 \ge \lambda_1$, the bound holds for all three.

Discretisation. The v_i are conforming P_2 functions on the $n = 48$ mesh of Section 7.2 (untrusted float eigenvectors, used as exact data). The σ_i lie in a Raviart–Thomas space of degree 2 in each reference simplex, with unknown $j = \sqrt{g}\,\sigma$. Its degrees of freedom are the outward flux densities per unit edge parameter $s \in \{0, \frac{1}{2}, 1\}$ and six interior moments. The flux through

an edge segment is the intrinsic 1-form $\iota_\sigma dA$, and the edge point $W = sP_i + (1-s)P_j$ is the same from both sides, also across the projective side pairings, since isometries are linear on W and preserve s. Equal and opposite edge degrees of freedom therefore make σ_i exactly $H(\mathrm{div})$-conforming on the glued surface, by the same mechanism as P_2 conformity. In the reference chart, $|\sigma|^2 dA = j^{\mathsf{T}}(LL^{\mathsf{T}} - QG)j/(D\sqrt{Q})\,da\,db$ and $(\operatorname{div}\sigma)\sqrt{g} = \partial_a j^a + \partial_b j^b$ with $\sqrt{g} = DQ^{-3/2}$. The entries of A_0, A_1, A_2 are sums over the 18 432 elements of integrals $\int p \cdot Q^{-s}$ ($s \in \{\frac{1}{2}, \frac{3}{2}\}$), enclosed as in Section 7.2 with 14 series terms and a Lagrange remainder. The σ_i are float minimisers of $b_G(w_i, w_i)$ (a sparse positive definite solve).

Proposition 7.4 (Certified)**.** *In Arb, $B \succ 0$ and $\bar{\nu}B - A \succ 0$ with $\bar{\nu} = -30.031942622361008 < 0$, by Sylvester's criterion. Hence $\nu_1, \nu_2, \nu_3 < \bar{\nu} < 0$, $q = 3$, and Theorem 7.3 with $k = 3$ gives*

$$\lambda_1(B) \geq 4 - \frac{5}{1-\bar{\nu}} \in [3.83887570105272435423062052218708564 07 \pm 6.06 \times 10^{-38}],$$

so $\lambda_1(B) \geq 3.8388757010527243$ (rounded down).

At $n = 24$ the same construction gives only 3.8387028952. The float ν at $n = 48$ are -30.03241, -30.03241 and -30.03194.

Checks. (i) On the flat torus $\mathbb{R}^2/\mathbb{Z}^2$ ($\lambda_1 = 4\pi^2$ with multiplicity 4; $n = 4$ trial functions, $\rho = 79.9 \leq 1 + 8\pi^2$) the same code certifies 39.2041077545, 39.4582495177, 39.4770992124 and 39.4783342462 on $N = 4, 8, 16, 32$ meshes, all below $4\pi^2 = 39.4784176044$ and converging at rate $O(h^4)$. (ii) The global Green identity $\sum_T \int (\sigma \cdot \nabla v + (\operatorname{div}\sigma)v) = 0$, evaluated in Arb for the certified fluxes, has residual at most 7.4×10^{-36}. (iii) Making the fluxes non-conforming (all edge signs $+1$) makes B indefinite on the torus, as the theory predicts. (iv) An independent reviewer redid the final step from the stored enclosures of A_0, A_1, A_2 in mpmath interval arithmetic and obtained the same bound. The reviewer also reproduced the element integrals from a from-scratch metric to relative accuracy 1.2×10^{-32}, checked every glued edge geometrically, and confirmed on an exact toy model that wrong indexing of the zero mode, or an invalid ρ, produces false bounds.

The hypothesis on ρ is not self-checking. With $\rho = 6.5$, or $\rho = 7.5$ at $n = 24$, both above $\lambda_4 + 1$, the computation still returns values below the true λ_1. A wrong ρ would therefore go undetected. The trace-formula certificate of Lemma 7.2(a) is load-bearing.

Scope. The interval $[3.8388757010, 3.8388872830]$ does not by itself imply multiplicity 3, so Lemma 7.2(c) and the certificates below still use [15, Lemmas 7.2–7.3].

7.4. From finite-element functions to exact eigenfunctions.

Lemma 7.5 (Gap transfer)**.** *Let $u \in H^1(B)$ with $m(u, 1) = 0$, let $e = P_E u$ and $w = u - e$, and let $\mu = q(u)/m(u)$. If $\lambda_{\mathrm{low}} \leq \lambda_1$ and $\lambda_4 \geq \Lambda_4 > \lambda_1$, then*

$$m(w) \leq \frac{\mu - \lambda_{\mathrm{low}}}{\Lambda_4 - \lambda_{\mathrm{low}}}\, m(u), \qquad q(w) \leq \frac{(\mu - \lambda_{\mathrm{low}})\Lambda_4}{\Lambda_4 - \lambda_{\mathrm{low}}}\, m(u).$$

Proof. w is orthogonal to $E \oplus \mathbb{R}1$ in both m and q, so $q(w) \geq \lambda_4 m(w)$ and $q(u) - \lambda_1 m(u) = q(w) - \lambda_1 m(w) \geq \max\{(\lambda_4 - \lambda_1)m(w), (1 - \lambda_1/\lambda_4)q(w)\}$. Since $\mu < \Lambda_4$, both right-hand sides decrease when λ_{low} is replaced by λ_1 or Λ_4 by λ_4, so the stated bounds follow. □

This is where the lower bound for λ_1 enters the certificates of Sections 7.5 and 7.6; we use $\Lambda_4 = 4.0$ and λ_1 in the ball

$$3.83888149202636215 \pm 5.791 \times 10^{-6} = [3.838875701026362,\ 3.838887283026362].$$

This ball contains $[3.8388757010527243,\ 3.8388872829261946]$, the interval given by Proposition 7.4 and the P_2 bound. The scripts differ from the original ones only in this ball and the output file name.

Alternative with Hypothesis 1.1. The original runs used the ball $3.8388872588 \pm 1.0000001 \times 10^{-6}$. That ball's lower end lies 4.2×10^{-11} below the lower end of Hypothesis 1.1 (conservative), and its upper end, 4.2×10^{-11} below that of Hypothesis 1.1, is covered by the certified P_2 bound $\lambda_1 \le 3.8388872830$. A re-run of the sign certificate with the exact lower end 3.83888625884 gives -12.850504 ($n = 96$).

7.5. **The certificate for $A \neq 0$.** The pulled-back forms along the deformation $x(t)$ of Proposition 5.2 are those of (1), and $b = q' - \lambda_1 m'$ is computed elementwise by forward-mode dual numbers over Arb, using $\frac{d}{dt}\int pQ^{-s} = \int \dot{p}Q^{-s} - s\int p\dot{Q}Q^{-s-1}$. Let x be the (untrusted) float P_1 eigencombination minimising b, and $u = x - (m(x,1)/m(1,1)) \cdot 1$, exactly m-orthogonal to constants. With $e = P_E u$ and $w = u - e$,

$$b(e,e) = b(u,u) - 2b(u,w) + b(w,w),$$

and $|b(u,w)| \le (\sum_T \beta_T^2 q_T(u))^{1/2} q(w)^{1/2} + \lambda_1 (\sum_T \beta_{\rho,T}^2 m_T(u))^{1/2} m(w)^{1/2}$, $|b(w,w)| \le \beta_{\max} q(w) + \lambda_1 \beta_{\rho,\max} m(w)$, where $-\beta_T a_T \le \dot{a}_T \le \beta_T a_T$ and $|\dot{\rho}_T| \le \beta_{\rho,T}\rho_T$ are enclosed in Arb over a parameter box containing the simplex. Here $\beta_T \ge \|N^{-1}\dot{N}\|_F + |\dot{D}/D + \dot{Q}/(2Q)|$ and $\beta_{\rho,T} \ge |\dot{D}/D - 1.5\,\dot{Q}/Q|$. Finally $m(e) = m(u) - m(w) > 0$, so $e \neq 0$. The results are in Table 5; at $n = 192$, $b(e,e) \le -3.0383476242423058 \cdot m(u)$ with the ball of Section 7.4 ($-3.0875694043934345 \cdot m(u)$ with the ball of Hypothesis 1.1). An independent re-run without the symmetry shortcut gives $-3.0383476316928864 \cdot m(u)$.

The printed $\beta_{\max} \approx 33$ is mostly enclosure overestimation (Frobenius norm and dependency over a box containing the simplex), not geometry: at the side-midpoint elements the sampled true supremum is about 3.3. Overestimation is safe. The certificate survives any λ_1 lower bound ≥ 3.83788; with $\lambda_4 \ge 3.9$ it still gives -1.89; and a mesh-only deformation, for which $A = 0$ exactly, is correctly not certified ($+1.409$).

Symmetry shortcut. The deformation and the mesh are invariant under the rotation by $\pi/2$, and the eight sectors are congruent blocks with identical local node order, so only two sectors are computed. This is justified by exact invariance and was checked against the full computation at $n = 48$ (identical output) and by an independent computation of all 4608 elements at $n = 16$ (55 296 pairs of entry enclosures, none non-overlapping).

7.6. **The sign certificate.** *Correctors (untrusted).* ψ_i solves the discrete first-order system $(K - \bar{\lambda}M)\psi_i = -(K' - \bar{\lambda}M')x_i$ on the M-complement of $\{x_1, x_2, x_3, 1\}$, then adds $-\frac{1}{2}\sum_j m'(x_i,x_j)x_j$ (so $M_1 \approx 0$) and $-m'(x_i,1)/m(1,1)$ (so $c_1 \approx 0$). These choices only sharpen the bound.

Jets. Everything is computed with order-2 Taylor jets in p over Arb: the vertices (F_B in closed form), the mesh nodes, $N = \mathrm{adj}(LL^{\mathsf{T}} - QG)$, D, Q and the integrals, using $Q(p)^{-s} = Q_0^{-s}[1 - s(Q_1/Q_0)p + (-sQ_2/Q_0 + \frac{s(s+1)}{2}(Q_1/Q_0)^2)p^2 + \cdots]$. Element derivatives agree with central differences to $O(h^2)$.

Transfer. Put $u_i = x_i - c_i 1$ with $c_i = m(x_i,1)/m(1,1)$ enclosed in Arb, $e_i = P_E u_i$ and $w_i = u_i - e_i$. Lemma 7.5 bounds w_i (Table 7). Every entry of X, A, M_1, c_1, M_0 at e is its value at u plus cross terms in w, bounded using elementwise relative bounds $-\beta^1 a \le \dot{a} \le \beta^1 a$, $-\beta^2 a \le \ddot{a} \le \beta^2 a$ (and similarly for ρ), enclosed over three boxes covering the simplex; quadratic terms in w use $\beta_{\max}$. The 3×3 algebra is done in Arb.

Result. At $n = 96$, with the ball of Section 7.4: $g''(0) \le -9.341154560446736$, central value $[-14.274 \pm 2.13 \times 10^{-4}]$ (an independent re-run gives -9.341154566965994). The distances to $E \oplus \mathbb{R}1$ are larger than with Hypothesis 1.1 (Table 7), because they are dominated by the width of the λ_1 ball.

Alternative with Hypothesis 1.1. At $n = 96$: $g''(0) \le -12.850476214196528$, central value $[-14.2742 \pm 2.23 \times 10^{-5}]$; the joint transfer error is about 1.42, so the margin is about nine times the error. The rest of this subsection refers to these runs. Table 6 lists the results for $n = 24, 48, 96$ and Table 8 the error budget. The breakdown rows of Table 8 are "one at a time" (each error group switched on alone); they do not add up to the joint error, and adding them to the central value would give about -12.864, which is *below* the certified value. Only the joint ball is a bound.

Robustness. The symmetry shortcut is not needed: runs without it certify at $n = 24$ and $n = 48$ (-12.566084126010532), and an independent re-run without it at $n = 96$ gives -12.850476213730866. The β values are enclosure values: on about 150 sampled elements the ratio of the observed to the enclosed value is at most 0.70 (β^1), 0.89 (β^2), 0.99 (β^1_ρ) and 0.97 (β^2_ρ).

Sensitivity to the λ_1 lower bound. The $n = 96$ certificate holds for every rigorous lower bound $\lambda_1 \ge 3.8388029653$ and fails at 3.8388029297. The bound of Proposition 7.4 clears this threshold by about 7.3×10^{-5}.

Negative controls. Each of the following fails to certify, as it should: $\psi = 0$; λ-radius 10^{-4}; $\lambda_4 = 3.84$; a λ-ball excluding the true eigenvalue.

Independent floating-point checks (not part of the proof). A separately written conforming P_1 code on a different mesh, with re-meshed finite differences, gives $\Lambda'' \approx -14.26$ by Richardson extrapolation at $n = 64/128$; the five-point difference of the discrete P_2 cluster sum gives -14.2739 at $n = 24$.

7.7. **Rank certificates.**

Proposition 7.6. *The 5×16 constraint Jacobian J at x_B has rank 5: its minor in columns $(0, 2, 5, 7, 8)$ has determinant $[46151927.034068685786828843022 58 \pm 2.89 \times 10^{-24}]$. For the regular octagon, $d(\text{angle sum})/dR = [-5.57297119598410437898199190923 6 \pm 5.18 \times 10^{-31}]$, in agreement with the closed form $8\, d\theta/dR$.*

Proof of Proposition 6.2. The pairings g_k are computed in Arb at 128 bits from the frame formula of Section 2.2, with derivatives by forward-mode dual numbers (no finite differences). At x_B, $|g_k V_{k+4} - V_{k+1}|, |g_k V_{k+5} - V_k| \le 1.2 \times 10^{-29}$, $|\mathsf{B} g_k^{\mathsf{T}} \mathsf{B} g_k - I| \le 4.4 \times 10^{-29}$, the constraints enclose 0, and $\operatorname{tr} g_k$ overlaps $1 + 2\cosh(2\operatorname{arccosh}(1+\sqrt{2})) = 22.3137084989848$, so each g_k translates by the systole. The six traces at x_B are 340.705627484771 ($g_0 g_1, g_1 g_2, g_2 g_3$) and 134.882250993909 ($g_0 G_2, g_1 g_3, g_1 G_3$). A basis of $\ker J$ is formed exactly in Arb from the pivot minor of Proposition 7.6, $T_b = [-J_{\text{piv}}^{-1} J_{\text{free}}; I]$, using the remaining 11 coordinates as free parameters. The resulting 6×11 Jacobian has 392 of its 462 maximal minors certified nonzero; the best uses tangent columns $(1, 3, 5, 7, 8, 10)$, i.e. the free coordinates $3, 6, 10, 12, 13, 15$, with determinant $[7872428136359854080.0 \pm 3 \times 10^{-6}]$. As a cross-check, an 11×11 minor of the stacked $[J; J_f]$ has determinant $[-1.3827971011809151404 \times 10^{24} \pm 2.74 \times 10^4]$. □

The four generator traces τ_{g_k} have rank only 3 on $T_{x_B}\mathcal{C}$, and several symmetric-looking six-word sets have rank 5 at B; these coincidences are forced by the octagon symmetry, which is why the specific words above are used. In an orthonormal tangent basis the six rows have singular values between 5.7×10^3 and 2.65×10^2 (floating point).

8. Proof of the Main Theorem

Proof. Item (i) is Lemma 7.2, and item (ii) is Corollary 5.3 with Section 4. Work in the chart $\mathcal{S}$ of Lemma 6.3.

(1) $\Lambda_{\mathcal{S}}$ is real-analytic, and $M_{\mathcal{S}}$ has the expansion of Theorem 3.1 with first-order term $A \circ \sigma$ (Lemma 3.3(a), Lemma 6.3(d)); $\operatorname{tr}(A \circ \sigma) \equiv 0$ (Proposition 5.1).

(2) $K_{\mathcal{S}} := \ker(A \circ \sigma) = \sigma^{-1}(K_0)$ is 3-dimensional (Corollary 5.3; σ is an isomorphism by Lemma 6.3(b)).
(3) $dp\, v_B \in K_{\mathcal{S}} \setminus \{0\}$, since $\sigma(dp\, v_B) = \sigma(v_B) = w_B \in K_0 \setminus \{0\}$ (Lemmas 4.7 and 6.3(c)), and $D^2\Lambda_{\mathcal{S}}(dp\, v_B, dp\, v_B) = D^2\Lambda_{\mathcal{C}}(v_B, v_B) \le -9.3411$ (Proposition 6.8, Lemma 6.3(c)). This is item (iv).
(4) By Lemma 6.4, $D^2\Lambda_{\mathcal{S}}|_{K_{\mathcal{S}}} = c \cdot \sigma^* g_{\mathrm{inv}}$. By step 3, $c\, g_{\mathrm{inv}}(w_B, w_B) \le -9.3411$, so $c < 0$ and $D^2\Lambda_{\mathcal{S}}|_{K_{\mathcal{S}}}$ is negative definite. This is item (iii), and hypothesis (b) of Lemma 6.1 holds with $\kappa > 0$.
(5) Lemma 6.1 gives $\lambda_1(\pi(x)) = \lambda_{\min} M_{\mathcal{S}}(x) < \lambda_*$ for $x \in \mathcal{S}$ near x_B, $x \neq x_B$. Case (i) of its proof gives the first-order statement of item (iii).
(6) Since $\pi|_{\mathcal{S}}$ is a homeomorphism onto a neighbourhood N of B (Lemma 6.3(a)), $\lambda_1 < \lambda_1(B)$ on $N \setminus \{B\}$. □

TABLE 1. External inputs. "Quoted" means a published statement used without re-proof.

Input	Used in	Status
Lehmann–Goerisch theorem [28, Thms. 4.1, 4.3]; second source [1, Thm. 2.1] as reproduced in [37, §2]	Proposition 7.4, hence Lemma 7.5	quoted from secondary sources; Goerisch's original papers not read
Selberg trace formula in the normalisation of [15, (3.1)] (from [4]); [15, Lemma 3.2] and its proof	Lemma 7.2(a)	quoted
Systole $2\operatorname{arccosh}(1+\sqrt{2})$ and multiplicities $\ge 12, 12, 24$ [15, §7.4]	Lemma 7.2(a), hence $\rho = 5$ in Proposition 7.4	quoted; lower bounds only
No 1- or 2-dimensional irreducibles in the λ_1-eigenspace [15, Lemmas 7.2–7.3]	Lemma 7.2(c) (not Proposition 7.4)	quoted
$\lvert\operatorname{Isom}(B)\rvert = 96$, $\operatorname{Isom}^+(B) \cong GL(2,3)$, octahedral quotient [15, §7.1]; finite $S_4 \subset PSL(2,\mathbb{C})$ is octahedral	Section 4	quoted / standard
$T_B\mathcal{T}_2 \cong \{\operatorname{Re}\varphi\}$ [29, 36, 8]	Lemma 3.3, Section 4	quoted
Fischer–Marsden splitting (as stated in [29, Rem. 3.3])	Lemma 3.3	quoted
Min-max ([7, Thm. 1.22], from Chavel); Ky Fan [30, 11, 12]	Lemmas 7.2, 6.6	quoted; Ky Fan reduced to min-max
Elliptic regularity, Rellich compactness, spectral theorem, Neumann series, Riemannian centre of mass in $\mathbb{H}^2$, Poincaré polygon theorem, invariance of domain	Sections 2–6	standard
$\mathcal{T}_2$ is a topological 6-manifold [13]; continuity of the extended mapping class group action	Lemmas 6.3, 6.4	quoted / standard
python-flint 0.9.0 (Arb), sympy 1.14.0	Section 7	trusted software
Optional, not used: Hypothesis 1.1 [34, §5]	cross-check; alternative constants	quoted

9. Formal verification in Lean

We formalised the finite-dimensional part of the argument in Lean 4 (toolchain v4.34.1, with Mathlib). The development has about 2250 lines in 8 files and contains no `sorry`, no `axiom`

declarations and no `native_decide`. Its main theorem is

$$\texttt{theorem Bolza.bolza_strict_local_max (I : BolzaInputs)}:$$
$$\forall^{\mathrm{f}} X \in \mathcal{N}(\texttt{I.B}),\quad X \neq \texttt{I.B} \;\Rightarrow\; \texttt{I.lam1}\, X < \texttt{I.lam1}\, \texttt{I.B},$$

where $\forall^{\mathrm{f}} X \in \mathcal{N}(\cdot)$ is Mathlib's "eventually in the neighbourhood filter" and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`.

9.1. **Scope.** Every analytic, geometric and computational input is a named field (an explicit hypothesis) of the structure `BolzaInputs`. These fields encode: $\mathcal{T}_2$ as a topological space and the chart of Lemma 6.3(a); the cluster matrix of Theorem 3.1 (C^2, symmetric, $M(0) = \lambda_* I$, $\lambda_1 \leq$ Rayleigh quotients); the representation of G on E and the equivariance of Lemma 3.3(b); irreducibility of E; the group relations used in Proposition 4.3; $A \neq 0$; the invariance of Lemma 6.4; the vector w_B and its symmetries; the curve F_B in the chart; the operator Ky Fan comparison; and the certified value of $g''(0)$.

Machine-checked from these inputs are: the local-maximum criterion (Lemma 6.1) in C^2 form, including a uniform second-order Taylor theorem; $\operatorname{tr} A \equiv 0$; $K_0 \subset \ker A$, $\ker A \subset K_0$ and $D^2\Lambda|_{K_0} = c\, I$; the exact line of Lemma 4.7; the step $c < 0$; the finite-dimensional Ky Fan theorem; the step $(\Lambda \circ \gamma)'' \leq g''$ from the Ky Fan comparison; and the second-order chain rule turning it into $D^2\Lambda(w_B, w_B) \leq -12.8504$. The exact representation theory (the 48-element image group, characters and multiplicities, generator matrices, change of basis) is checked by kernel evaluation. The final arithmetic of each certificate (e.g. $0.912612 < 1$, $3.869661 < 4$, $-12.850476214196528 \leq -12.8504$, the determinant enclosures excluding 0) is checked by `norm_num`. A toy instance of `BolzaInputs` shows that the hypotheses are jointly satisfiable; it says nothing about B.

9.2. **Limits.** The Lean development does *not* verify: that the fields of `BolzaInputs` hold for the Bolza surface (this is the content of Theorem 3.1, Lemmas 3.3, 6.3 and 6.4, the certificates, and the quoted results); the Arb computations themselves; the operator min-max and Ky Fan inequality on H^1; or the identification of the exact matrices with the geometric action on TT tensors. The step from character norm 1 to absolute irreducibility is not formalised; instead, the facts actually used (Schur-type invariant forms on K_0, the absence of invariant subspaces in W_0, and the involution argument) are proved directly over $\mathbb{R}$.

The Lean development takes the SU-free constant -9.3411 of Proposition 6.8 as its hypothesis on $g''(0)$; the constant -12.8504 obtained under Hypothesis 1.1 is recorded as an alternative that discharges the same hypothesis. The argument uses the constant only through $c < 0$. The Lehmann–Goerisch certificate that produces -9.3411 is checked by the independent certificate checker, not in Lean.

9.3. **Observations from the formalisation.**

(1) The hypothesis $A \neq 0$ is not needed for Lemma 6.1; symmetry of M is used only to pass to an eigenvalue statement. The constant c_A comes from an elementary bound: a traceless symmetric 3×3 matrix S has a test vector with entries in $\{0, \pm 1\}$ and $x^{\mathsf{T}} S x \leq -\|S\|_{\sup}/4$.
(2) $\operatorname{tr} A \equiv 0$ needs equivariance under r only.
(3) Both $\ker A = K_0$ and the scalar form of the Hessian genuinely use the non-octagon element t (Section 4.3).
(4) As noted after Table 8, the one-at-a-time error rows of the sign certificate are not additive; only the joint enclosure is checked.

10. Remarks and open problems

Remark 10.1 (The input from [34], and the weakest inputs). An earlier version of the proof depended on Hypothesis 1.1. It rests on a single sentence of [34, §5] reporting a rigorous implementation whose code is unpublished, and the sentence does not say whether the error 10^{-6} is absolute or relative. Proposition 7.4 removes this dependence. Hypothesis 1.1 remains consistent with our enclosure $[3.8388757010, 3.8388872830]$. If it is assumed, the constants improve to -12.8504 (second order) and $-3.0875 \cdot m(u)$ (first order). The second-order certificate needs only $\lambda_1(B) \geq 3.8388029653$; under Hypothesis 1.1 this leaves 83 times its stated error, and it still holds if that error is read as relative ($n = 48$, radius 10^{-5}, gives -9.43).

The weakest inputs are now quoted results rather than computations: [15, Lemmas 7.2–7.3] (for multiplicity 3), the geodesic data of [15, §7.4] with the trace formula (for $\lambda_4 > 4.0$, which is the load-bearing hypothesis $\rho = 5$ of Proposition 7.4), and the Lehmann–Goerisch theorem. We quote the latter from secondary sources [28, 37] (the second reproducing [1]); we have not read Goerisch's original papers. A sharper lower bound is available by refining the mesh ($n = 96$ was not run). Other routes would be Liu-type projection bounds, nonconforming elements in the spirit of [6] (whose interpolation estimate does not directly apply to projectively glued edges), or an a posteriori bound for the method of particular solutions.

Remark 10.2 (Strohmaier's mechanism). Off K, case (i) of Lemma 6.1 shows that λ_1 drops linearly, and since $A(w)$ is traceless and nonzero the three branches split to first order; this is consistent with the statement of [35] that no branch is extremal. On K the first-order term vanishes, all three branches are stationary, and λ_1 has a strict quadratic maximum. Numerically, along F_B we observe $\lambda_1 = \lambda_1(B) - 21.2p^2 + O(p^4)$ with $\lambda_2 = \lambda_3$ having a minimum, while along the equilateral family F_A (tangent to δ) $\lambda_1 = \lambda_1(B) - 5.715|p| + O(p^2)$. These values are numerical (uncertified finite-element computations) and are not used in the proof. The coincidence $\lambda_2 \equiv \lambda_3$ along F_B is also only observed numerically; the stabiliser of w_B in G has order 16 and contains the involution $u\colon x \mapsto i/x$, which suggests an explanation, but we have not completed it.

Remark 10.3 (Global maximality). The Main Theorem does not bear on global maximality, which remains open. Combining the rigorous bound of [27] with Proposition 7.4 shows only that no genus-two surface exceeds $\lambda_1(B)$ by more than $3.8388976481 - 3.8388757010$ (about 1.1×10^{-5} with Hypothesis 1.1), and [33] indicates that the existing single-correlator bootstrap will not close this gap. A quantitative version of our result, giving an explicit neighbourhood N, together with an improved global bound, could in principle reduce the global problem to a compact region away from B. We have not attempted this.

Remark 10.4 (Further questions).

(1) Is B a non-degenerate local maximum in the sharper sense $\lambda_{\min} Q_2|_K < 0$, and what is the second-order behaviour of the individual branches on K?
(2) The method uses only: a symmetric surface whose λ_1-cluster is isolated, symmetry forcing $\operatorname{tr} A \equiv 0$, an exact identification of $\ker A$, irreducibility of $\ker A$, and one certified second-order direction. Fortier Bourque and Petri suggest that m_1 should be maximised among local maximisers of λ_1 [14, §9]; it would be natural to apply the same method to other conjectured maximisers, such as the Klein quartic in genus three [15, 14].
(3) Can the remaining analytic inputs (Theorem 3.1, Lemma 3.3) and the Arb certificates be brought into the formal verification?

Acknowledgements. The computations, proofs and reviews in this work were produced with Claude (Anthropic) AI agents. Each component was independently re-checked by separate agent reviewers, who re-ran the key certificates and re-derived the main formulas. A check by a human expert is pending. The author is responsible for any remaining errors.

Appendix A. Certificate data

All values are copied from the output files listed in Appendix B. Enclosures $[m \pm r]$ are Arb balls.

Table 2. Trace formula counts (Lemma 7.2(a)); $N = 21$.

Run	s	Window	$c \geq$	Certified bound
A (Lemma 7.1)	1	$[0, 3.3]$	0.9816424759937855	$[0.912612352675290375 \pm 9.95 \times 10^{-20}] < 1$
B ([15, Lemma 3.2])	13/10	$[3.3, 4.0]$	0.9964283521820437	$[3.86966111976973759 \pm 1.65 \times 10^{-18}] < 4$
sanity	13/10	$[3.3, 3.9]$		3.4803 (≥ 3)
sanity	13/10	$[3.3, 5.4]$		7.5466 (≥ 7)

Table 3. Certified P_1 Rayleigh–Ritz bounds at B.

n	certified $\lambda_3(B) \leq$	area/4π
48	3.843638188137345	$[1.000000 \pm 1.50 \times 10^{-8}]$
96	3.8400767924773938	$[1.000000000 \pm 5.73 \times 10^{-11}]$
192	3.839184804702046	$[1.00000000000 \pm 2.23 \times 10^{-13}]$

Table 4. Certified P_2 Rayleigh–Ritz bounds at B.

n	dofs	certified $\lambda_3(B) \leq$	certified $\lambda_1 + \lambda_2 + \lambda_3 \leq$	area/4π
24	9 214	3.838893606899935	11.516677957238421	$[1 \pm 2.21 \times 10^{-10}]$
48	36 862	3.8388876433743824	11.516662767334967	$[1 \pm 1.16 \times 10^{-14}]$
96	147 454	3.8388872829261946	11.516661838577997	$[1 \pm 1.19 \times 10^{-18}]$

Table 5. Certificate for $A \neq 0$ (per unit t, $m(u)$ normalised to 1). The $n = 48$ row fails to certify.

n	$\mu = R(u)$	$b(u,u)$	$m(w) \leq$	$q(w) \leq$	transfer error	$b(e,e) \leq$
48	3.8420704	-4.808678	0.0198	0.0790	9.08	$+4.27$
96	$[3.8396833301 \pm 4.94 \times 10^{-11}]$	$[-4.806689 \pm 1.39 \times 10^{-7}]$	0.00494	0.0198	3.78	-1.0282409843057394
192	$[3.839086294670 \pm 5.19 \times 10^{-13}]$	$[-4.806192 \pm 1.96 \times 10^{-7}]$	0.0012	0.0049	1.71	-3.0875694043934345
SU-free ball (Section 7.4); the rows above use the ball of Hypothesis 1.1						
96	$[3.8396833301 \pm 4.94 \times 10^{-11}]$	$[-4.806689 \pm 8.00 \times 10^{-7}]$	$[0.0050 \pm 5.95 \times 10^{-5}]$	$[0.020 \pm 2.38 \times 10^{-4}]$	$[3.8 \pm 0.0246]$	-0.9989201873540878
192	$[3.839086294670 \pm 5.19 \times 10^{-13}]$	$[-4.806193 \pm 8.81 \times 10^{-7}]$	$[0.0013 \pm 6.49 \times 10^{-5}]$	$[0.005 \pm 2.29 \times 10^{-4}]$	$[1.7 \pm 0.0679]$	-3.0383476242423058

TABLE 6. Sign certificate along F_B (P_2).

n	dofs	central $g''(0)$	certified $g''(0) \le$	transfer error
24	9 214	-14.2738	-9.7138	4.56
48	36 862	-14.2742	-12.566084055230016	1.71
96	147 454	$[-14.2742 \pm 2.23 \times 10^{-5}]$	-12.850476214196528	1.42
96 (SU-free ball)	147 454	$[-14.274 \pm 2.13 \times 10^{-4}]$	-9.341154560446736	

The first three rows use the ball of Hypothesis 1.1; the last row uses the ball of Section 7.4 and is the one used in the proof. The rows use the $\pi/2$-symmetry shortcut. Without it: $n = 24$ certifies with the same result, $n = 48$ gives -12.566084126010532, and an independent re-run at $n = 96$ gives -12.850476213730866.

TABLE 7. Certified distances to $E \oplus \mathbb{R}1$ at $n = 96$ (Lemma 7.5).

i	$\mu_i = R(u_i)$	$\sqrt{q(w_i)} \le$	$\sqrt{m(w_i)} \le$
1	$[3.838887277826 \pm 2.34 \times 10^{-13}]$	5.0299×10^{-3}	2.515×10^{-3}
2	$[3.8388872778258 \pm 8.95 \times 10^{-14}]$	5.0299×10^{-3}	2.515×10^{-3}
3	$[3.838887282926 \pm 2.30 \times 10^{-13}]$	5.0425×10^{-3}	2.521×10^{-3}
SU-free ball (used in the proof)			
1	$[3.838887277826 \pm 2.34 \times 10^{-13}]$	0.016953498968304958	0.008476749477446675
2	$[3.8388872778258 \pm 8.95 \times 10^{-14}]$	0.016953498968304958	0.008476749477446675
3	$[3.838887282926 \pm 2.30 \times 10^{-13}]$	0.016957233113081952	0.00847861654983665

The first three rows use the ball of Hypothesis 1.1. There $\mu_i - \lambda_{\text{low}} \approx 1.02 \times 10^{-6}$ is almost entirely the width of the λ_1 interval (the discretisation part is about 2×10^{-8}). With the wider SU-free ball the distances are correspondingly larger. In both cases they cannot be reduced by refining the mesh, only by a sharper lower bound for λ_1.

TABLE 8. Error budget of the sign certificate at $n = 96$, for the run with the ball of Hypothesis 1.1 (alternative constant; the proof uses -9.341154560446736). Rows below the central value are one-at-a-time contributions (upper end minus central value with only that group switched on); they are not additive.

Term	Value
$T_2 = \operatorname{tr}(q'' - \lambda_* m'')(u, u)$ (float)	45.635701833848785
diagonal (certified)	[15.613361, 15.613361, 14.408980]
coupling (ψ terms, float)	-59.9099
$(q - \lambda_* m)(\psi_i)$	0.797, 0.797, 28.361
central $g''(0)$	$[-14.2742 \pm 2.23 \times 10^{-5}]$
X_1: $4\,b(\psi, w)$ cross terms	$+0.839368$
X_2: $2(q'' - \lambda_* m'')(u, w)$ and w–w terms	$+0.570060$
M_0	$+4.6 \times 10^{-4}$
A, M_1, c_1	1.5×10^{-5}, 1.5×10^{-5}, 7.4×10^{-5}
certified upper bound (joint ball)	-12.850476214196528

Enclosure maxima at $n = 96$: $\beta^1_{\max} = 13.2$, $\beta^2_{\max} = 38.1$, $\beta^1_\rho, \beta^2_\rho \le 1.49, 1.35$; energy-weighted rms values 4.2/9.1 for u and 4.6 (ψ_1, ψ_2), 4.0 (ψ_3) for ψ.

TABLE 9. Rank certificates.

Quantity	Certified value
5×5 minor of J, columns $(0, 2, 5, 7, 8)$	$[46151927.03406868578682884302258 \pm 2.89 \times 10^{-24}]$
d(angle sum)$/dR$, regular octagon	$[-5.57297119598410437898199190923 6 \pm 5.18 \times 10^{-31}]$
$\operatorname{tr} g_k$, $k = 0, \ldots, 3$	$[22.3137084989848 \pm 3.97 \times 10^{-14}]$
nonzero 6×6 minors of the trace Jacobian	392 of 462
best minor, tangent columns $(1, 3, 5, 7, 8, 10)$	$[787242813635985408 0.0 \pm 3 \times 10^{-6}]$
11×11 cross-check minor of $[J; J_f]$	$[-1.3827971011809151404 \times 10^{24} \pm 2.74 \times 10^{4}]$

TABLE 10. Exact characters (Proposition 4.1 and Section 4).

	r	r^2	J	t	s_{mid}	s_{vert}
T_B	-2	-2	6	0	0	0
K_0	-1			0	1	-1
W_0	-1			0	-1	1

$\langle \chi_T, \chi_T \rangle = 2$, $\langle \chi_T, 1 \rangle = 0$; K_0, W_0 have norm 1 and are orthogonal; χ_T vanishes on all orientation-reversing elements. Group: order 96; G^+ element orders 1:1, 2:13, 3:8, 4:6, 6:8, 8:12; D_8 contains 4 reflections of s_{mid}-type and 4 of s_{vert}-type.

Appendix B. Reproducibility

The scripts and the working record come from the private research archive in which this proof was developed (not public) and are not in the `bolzakit` repository. What the repository contains is `reproduce.py` and the `certificates/` directory next to this paper, which regenerate and check the certified computations. The original scripts were run with a virtual environment (python-flint 0.9.0, sympy 1.14.0, numpy, scipy), under `nice -n 10` and `OMP_NUM_THREADS=2`. Times are wall-clock on a shared laptop.

Command	Output	Time
`python lp_count.py A 3.3 21 1.0`	`lp_count.out`, `lp_A_3.3.json`	~ 3 s
`python lp_count.py B 3.3 4.0 21 1.3`	`lp_B_3.3_4.0.json`	~ 3 s
`python cert_rr.py 48 96 192`	`cert_rr_bolza.out`	~ 8 min
`python cert_rr_p2.py 24 48 96`	`cert_rr_p2_bolza.out`	~ 100 s
`SYM=1 python cert_first_order.py 96 192`	`cert_first_order.out`	~ 12 min
`python sym_check_elements.py 16`	`sym_check_elements.out`	~ 1 min
`python cert_rank.py`	`cert_rank.out`	~ 5 s
`python cert_dpi_rank.py`	`cert_dpi_rank.out`, `.json`	< 1 s
`python cert_chars.py`	`cert_chars_A.out`, `.json`	~ 22 s
`python cert_chars.py B 48 96` (not used in the proof)	`cert_chars_B.out`	~ 4.4 min
`python cert_sign.py 48 96`	`cert_sign.out`, `cert_sign_48_96.json`	~ 4.3 min
`SYM=0 python cert_sign.py 12 24`	`cert_sign_nosym.out`	
`SYM=0 python cert_sign.py 48`	`cert_sign_48_nosym.out`	
`python lb_torus.py` (validation)	`lb_torus.out`	
`python lb_bolza.py 48`	`lb_bolza_48.out`, `.json`	211 s
`python lb_cert_sign.py 96`	`lb_cert_sign.out`, `lb_cert_sign_96.json`	165 s
`python lb_cert_first_order.py 96 192`	`lb_cert_first_order.out`, `lb_cert_first_order_96_192.json`	195 s + 718 s

The core floating-point geometry is in `octagon.py`. The Lean development is in `lean/`; it is built by `lean/build.sh`, which compiles the files `RepTheory`, `LocalMax`, `Taylor`, `RepReal`, `Certificates`, `KyFan`, `Assembly` and `Model` with plain `lean` against prebuilt Mathlib `.olean` files and then prints the axioms of the main declarations (a clean build took 2 min 33 s).

Process. The certificates and written proofs were produced in an AI-assisted workflow (see the acknowledgements). Each component was re-checked by a separate reviewer: the trace-formula, P_1, $\operatorname{tr} A$ and $A \neq 0$ certificates; the sign certificate (including an independent re-derivation of Lemma 6.7, an independent P_1 code, sensitivity analysis and negative controls); the representation theory (with an independently written model); the proof of Lemma 3.3; the chart certificate and Lemma 6.4, whose certificate was re-run with bit-identical output; and the Lehmann–Goerisch lower bound (Proposition 7.4) with the re-certified first- and second-order certificates. That reviewer checked the quoted theorem against both secondary sources, redid the final step in independent interval arithmetic, re-ran the $n = 24$ and torus cases, and re-ran both downstream certificates. The review records are kept with the project files.

References

1. Henning Behnke, Ulrich Mertins, Michael Plum, and Christian Wieners, *Eigenvalue inclusions via domain decomposition*, Proc. R. Soc. Lond. A **456** (2000), 2717–2730.

2. J. Bonifacio, *Bootstrapping closed hyperbolic surfaces*, J. High Energy Phys. **03** (2022), 093, arXiv:2111.13215 (v2, 2025).
3. James Bonifacio, *Bootstrap bounds on closed hyperbolic manifolds*, J. High Energy Phys. **02** (2022), 025, arXiv:2107.09674.
4. Peter Buser, *Geometry and spectra of compact Riemann surfaces*, Modern Birkhäuser Classics, Birkhäuser, Boston, 2010.
5. Chunhao Cai, *Response calculus for spectral simplicity and joint eigenvalue densities*, arXiv:2608.02459 (v2), 2026.
6. Carsten Carstensen and Joscha Gedicke, *Guaranteed lower bounds for eigenvalues*, Math. Comp. **83** (2014), 2605–2629.
7. J. Cook, *Properties of eigenvalues on Riemann surfaces with large symmetry groups*, Ph.D. thesis, Loughborough University, 2018, hdl:2134/36294; arXiv:2108.11825.
8. Georgios D. Daskalopoulos and Richard A. Wentworth, *Harmonic maps and Teichmüller theory*, Handbook of Teichmüller Theory, Vol. I, IRMA Lect. Math. Theor. Phys., vol. 11, Eur. Math. Soc., 2007, arXiv:math/0604403, pp. 33–109.
9. Joaquim Duran, *The $\bar{\partial}$-Robin Laplacian*, J. Math. Anal. Appl. **558** (2026), no. 1, 130411, arXiv:2507.16895.
10. Ahmad El Soufi and Saïd Ilias, *Laplacian eigenvalue functionals and metric deformations on compact manifolds*, J. Geom. Phys. **58** (2008), no. 1, 89–104, arXiv:math/0701777.
11. K. Fan, *On a theorem of Weyl concerning eigenvalues of linear transformations I*, Proc. Nat. Acad. Sci. U.S.A. **35** (1949), 652–655.
12. ———, *On a theorem of Weyl concerning eigenvalues of linear transformations II*, Proc. Nat. Acad. Sci. U.S.A. **36** (1950), 31–35.
13. Anna Felikson and Sergey Natanzon, *Moduli via double pants decompositions*, arXiv:1102.4861 (v1), 2011.
14. M. Fortier Bourque and B. Petri, *Linear programming bounds for hyperbolic surfaces*, arXiv:2302.02540 (v3, February 2026), 2023.
15. ———, *The Klein quartic maximizes the multiplicity of the first positive eigenvalue of the Laplacian*, J. Differential Geom. **128** (2024), 521–556, arXiv:2111.14699.
16. Maxime Fortier Bourque, Dídac Martínez-Granado, and Franco Vargas Pallete, *The extremal length systole of the Bolza surface*, Ann. Henri Lebesgue **7** (2024), 1409–1455, arXiv:2105.03871.
17. Maxime Fortier Bourque and Kasra Rafi, *Local maxima of the systole function*, J. Eur. Math. Soc. **24** (2022), no. 2, 623–668.
18. Elliott Gesteau, Sridip Pal, David Simmons-Duffin, and Yixin Xu, *Bounds on spectral gaps of hyperbolic spin surfaces*, arXiv:2311.13330, 2023.
19. William Hide and Bram Petri, *Limit points of uniform arithmetic bass notes*, Compositio Math. **162** (2026), 1487–1506, arXiv:2412.15111.
20. D. Jakobson, M. Levitin, N. Nadirashvili, N. Nigam, and I. Polterovich, *How large can the first eigenvalue be on a surface of genus two?*, Int. Math. Res. Not. (2005), no. 63, 3967–3985, arXiv:math/0509398.
21. F. Jenni, *Über den ersten Eigenwert des Laplace-Operators auf ausgewählten Beispielen kompakter Riemannscher Flächen*, Comment. Math. Helv. **59** (1984), 193–203.
22. V. Kalvin, *Spectral determinants of the Bolza surface and the Klein quartic*, arXiv:2608.01611, 2026.
23. Chiu-Yen Kao, Rongjie Lai, and Braxton Osting, *Maximization of Laplace–Beltrami eigenvalues on closed Riemannian surfaces*, ESAIM Control Optim. Calc. Var. **23** (2017), no. 2, 685–720, arXiv:1405.4944.
24. Mikhail Karpukhin, *Second variation formula for eigenvalue functionals on surfaces*, arXiv:2508.10465, 2025.
25. Mikhail Karpukhin, Romain Petrides, and Daniel Stern, *Existence of metrics maximizing the first Laplace eigenvalue on closed surfaces*, arXiv:2505.05293, 2025.
26. T. Kato, *Perturbation theory for linear operators*, Springer, 1995.
27. Petr Kravchuk, Dalimil Mazáč, and Sridip Pal, *Automorphic spectra and the conformal bootstrap*, Comm. Amer. Math. Soc. **4** (2024), 1–63, arXiv:2111.12716.
28. Xuefeng Liu and Michael Plum, *A two-stage finite element approach for high-precision guaranteed lower eigenvalue bounds*, arXiv:2512.23182, 2025.
29. Filippo Mazzoli, *The infimum of the dual volume of convex co-compact hyperbolic 3-manifolds*, Geom. Topol. **27** (2023), 2319–2346, arXiv:2101.09380.
30. Mohammad Sal Moslehian, *Ky Fan inequalities*, arXiv:1108.1467.
31. Shin Nayatani and Toshihiro Shoda, *Metrics on a closed surface of genus two which maximize the first eigenvalue of the Laplacian*, C. R. Math. Acad. Sci. Paris **357** (2019), no. 1, 84–98, arXiv:1704.06384.
32. Bram Petri, *Extremal and random hyperbolic geometry*, Habilitation thesis, Institut de Mathématiques de Jussieu–Paris Rive Gauche, 2024.

33. Alex Radcliffe, *Non-saturation of bootstrap bounds by hyperbolic orbifolds*, arXiv:2404.14479 (v2, November 2025), 2024.
34. A. Strohmaier and V. Uski, *An algorithm for the computation of eigenvalues, spectral zeta functions and zeta-determinants on hyperbolic surfaces*, Comm. Math. Phys. **317** (2013), 827–869, arXiv:1110.2150.
35. Alexander Strohmaier, *Computation of eigenvalues, spectral zeta functions and zeta-determinants on hyperbolic surfaces*, Geometric and Computational Spectral Theory, Contemp. Math., vol. 700, Amer. Math. Soc., 2017, arXiv:1604.02722, pp. 177–205.
36. Anthony J. Tromba, *Teichmüller theory in Riemannian geometry*, Birkhäuser, Basel, 1992.
37. Tomáš Vejchodský, *Flux reconstructions in the Lehmann–Goerisch method for lower bounds on eigenvalues*, J. Comput. Appl. Math. **340** (2018), 676–690, arXiv:1801.02561.

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